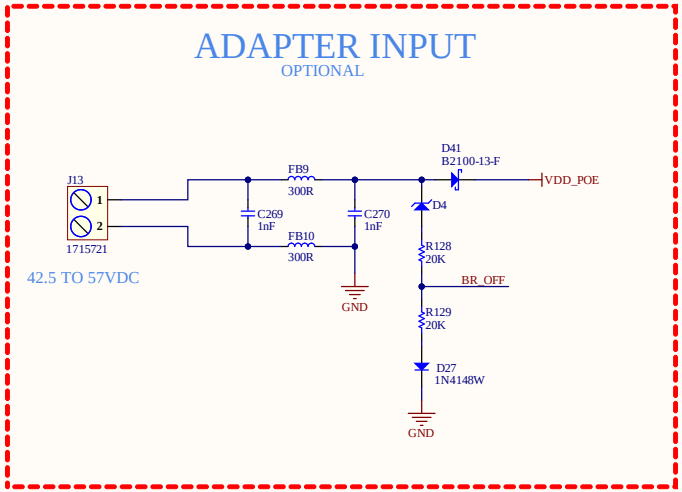
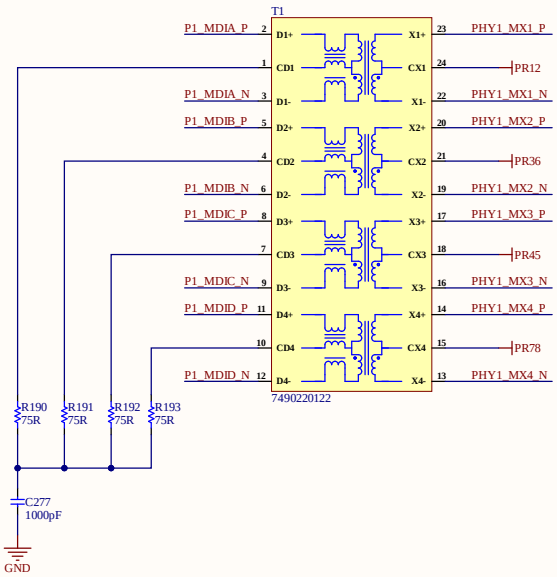
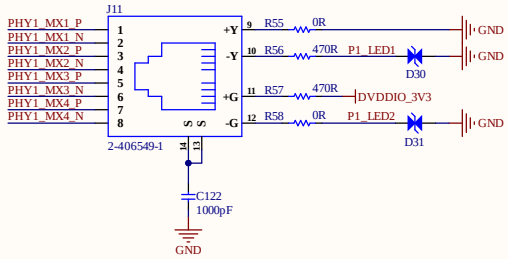
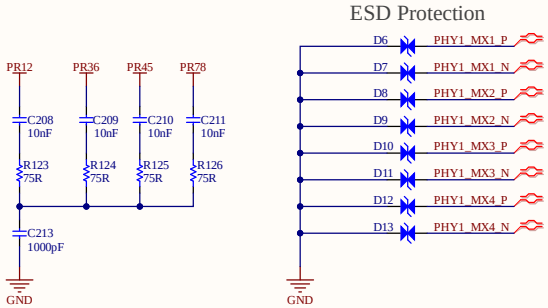
ETHERNET_PoE PD INTERFACE

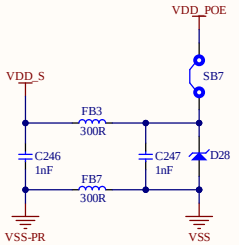
POE Power Device

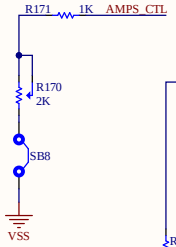


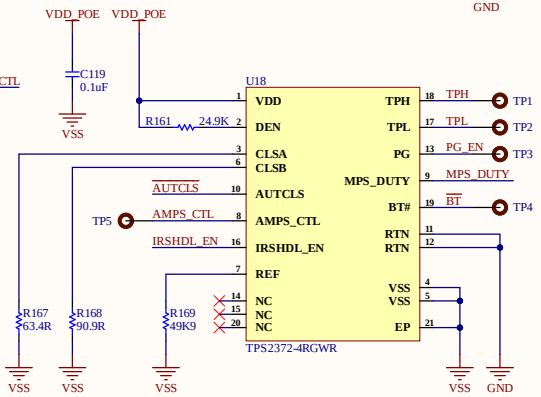


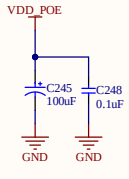


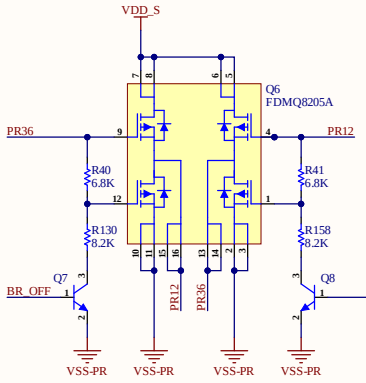


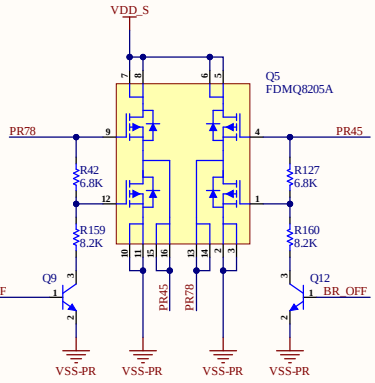


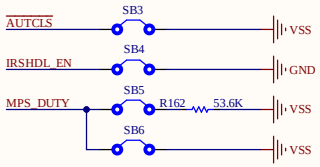


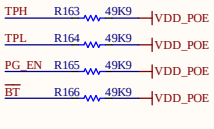


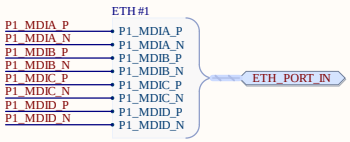


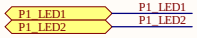


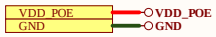










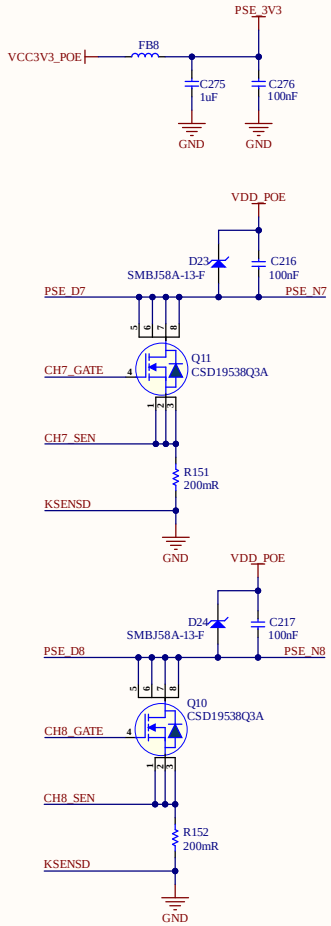


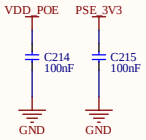
eSpark Consulting Group

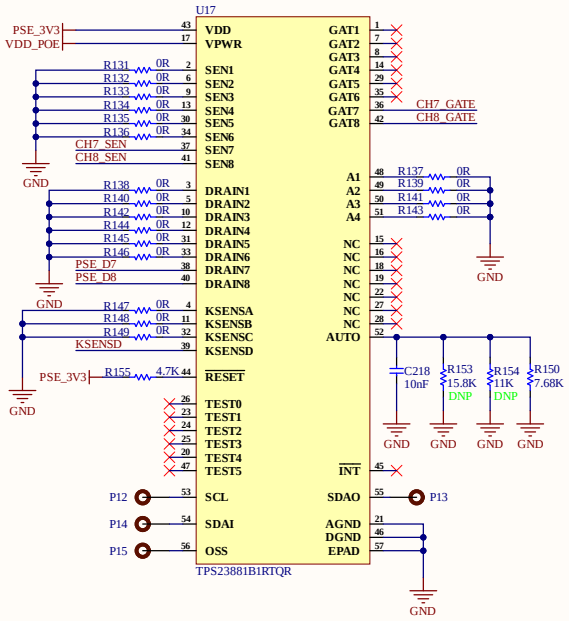
Title: ETH_PoE-PD_INTERFACE.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 1
Drawn By: Huzaifa Ali	Page	Total: 27

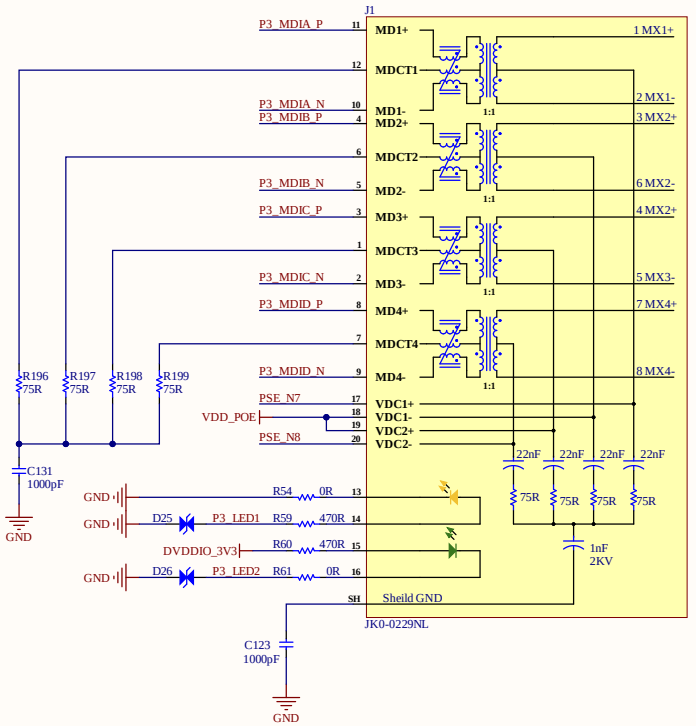
PoE_PSE_CONTROLLER

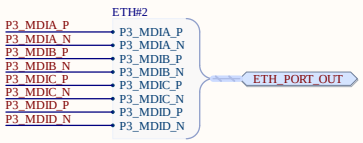
Power sourcing equipment

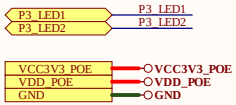








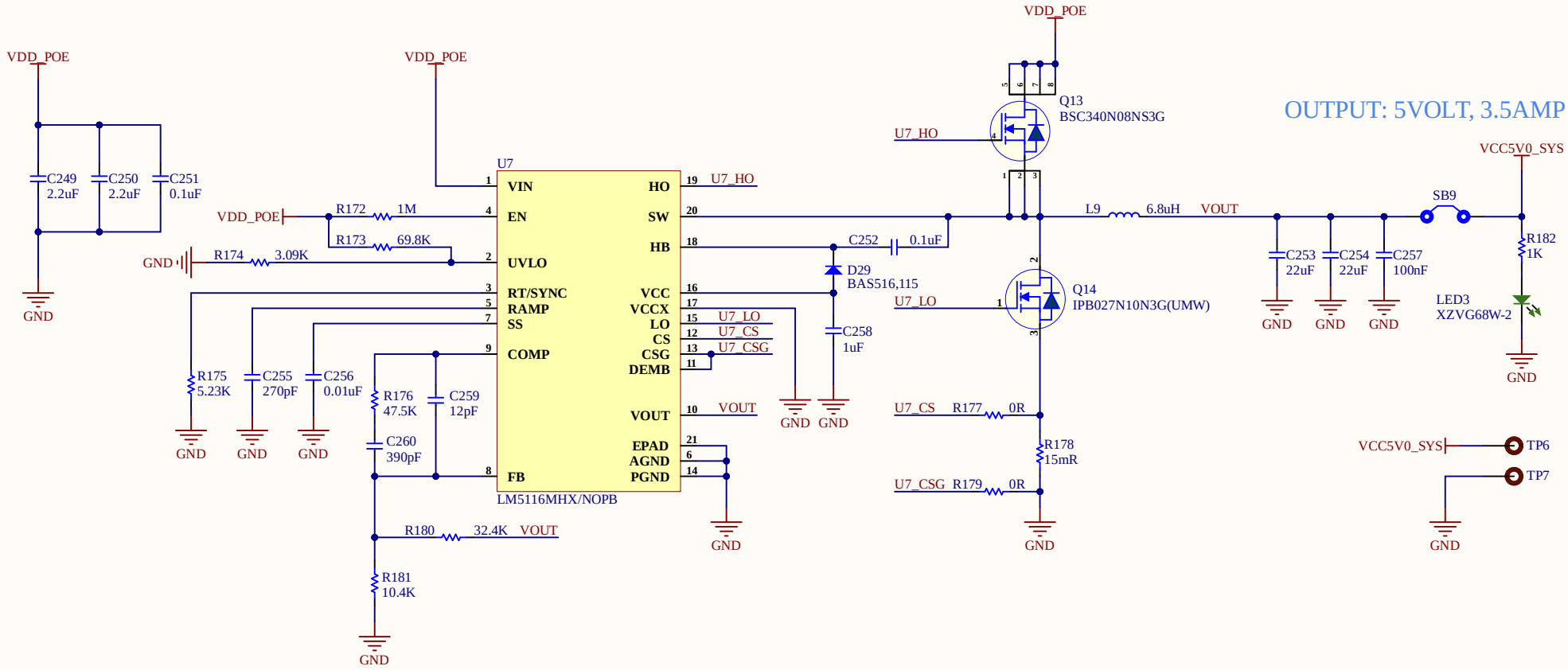




eSpark Consulting Group

Title: PoE_PSE_CONTROLLER.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 2
Drawn By: Huzaifa Ali	Page	Total: 27

SYNCHRONOUS _BUCK_CONTROLLER

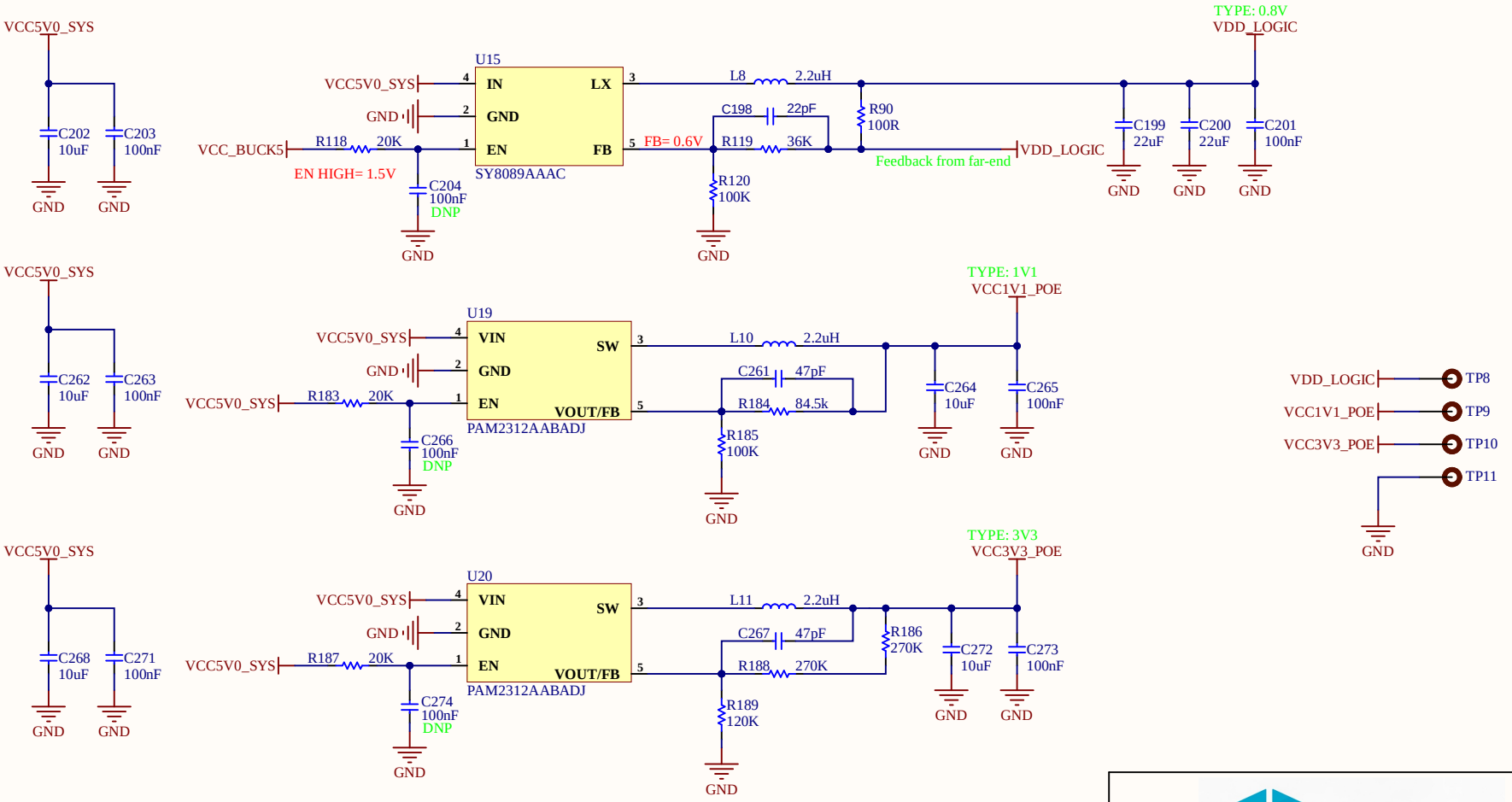


VCC5V0_SYS → VCC5V0_SYS
VDD_POE → VDD_POE
GND → GND

eSpark Consulting Group

Title: BUCK_CONTROLLER_SHEET.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 3
Drawn By: Huzaifa Ali	Page	Total: 27

POWER_LDO_SHEET

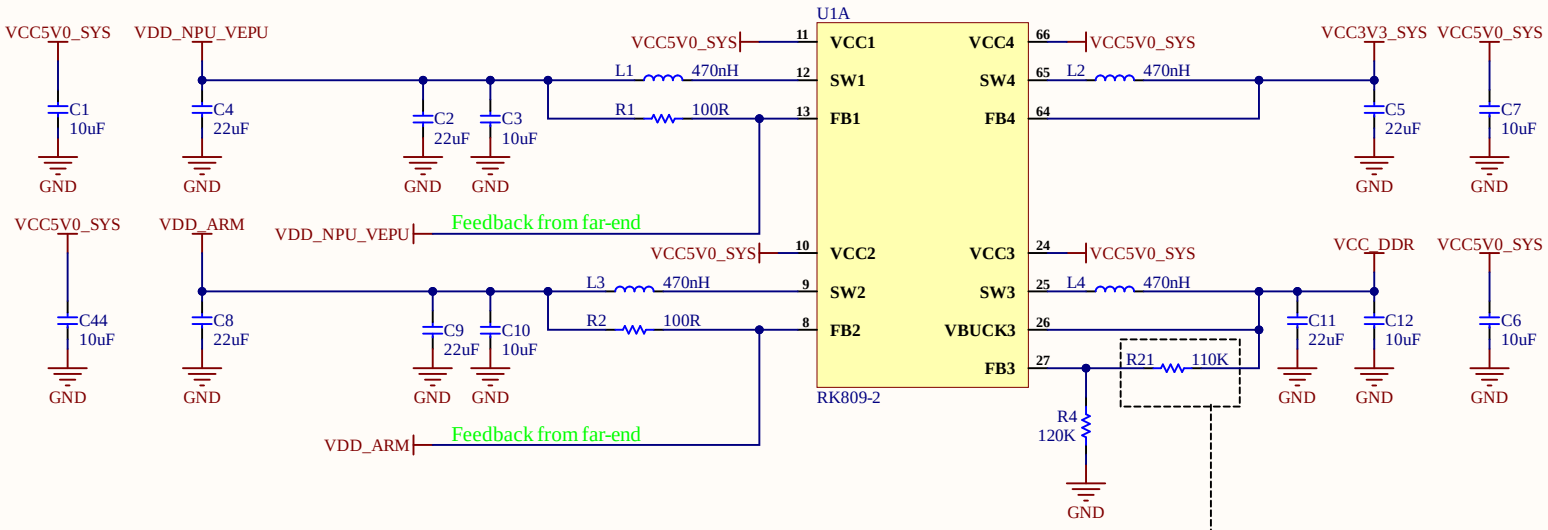


VCC3V3_POE	VCC3V3_POE
VCC1V1_POE	VCC1V1_POE
VCC5V0_SYS	VCC5V0_SYS
VDD_LOGIC	VDD_LOGIC
VCC_BUCK5	VCC_BUCK5
GND	GND

eSpark Consulting Group

Title: POWER_LDO_SHEET.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 4
Drawn By: Huzaifa Ali	Page	Total: 27

PMIC RK809 DC-DC



DDR Type	VCC_DDR Voltage	R21
DDR4	1.21V	62K
LPDDR3	1.21V	62K
DDR3L	1.35V	82K
DDR3(default)	1.53V	110K
LPDDR4	1.14V	51K

VCC5V0_SYS
VCC3V3_SYS
VDD_ARM
VDD_NPU_VEPU
VCC_DDR
GND

VCC5V0_SYS
VCC3V3_SYS
VDD_ARM
VDD_NPU_VEPU
VCC_DDR
GND

eSpark Consulting Group

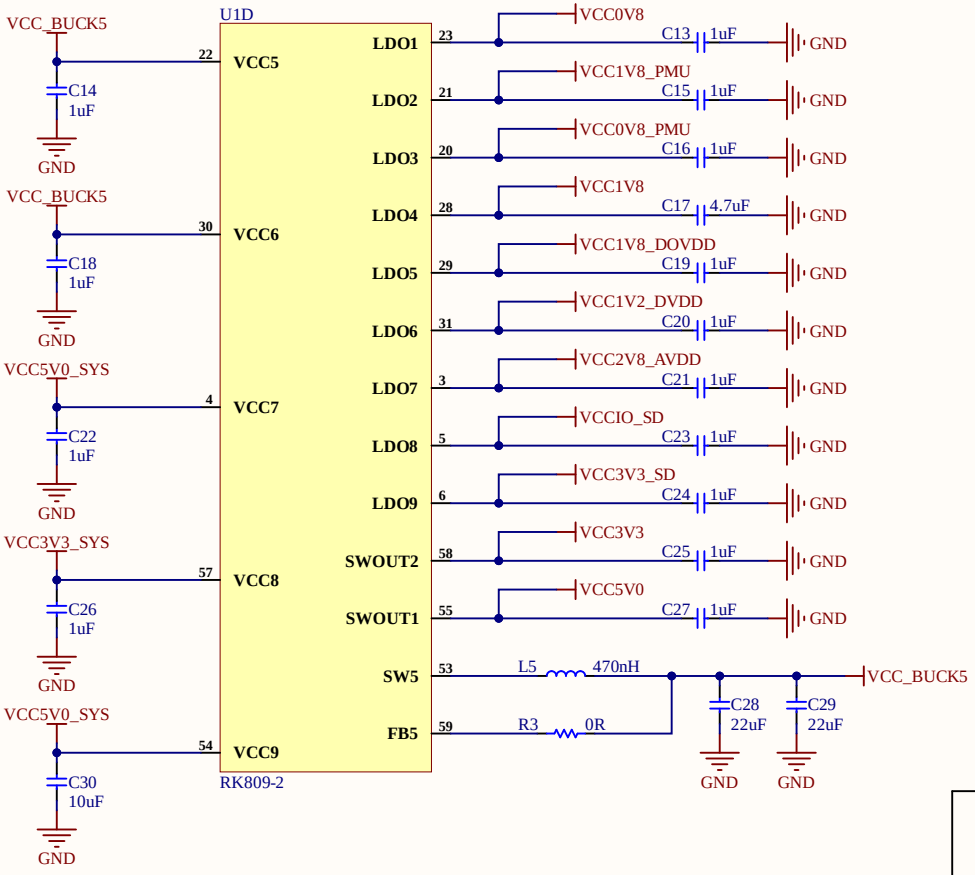
Title: POWER_PMIC_RK809_DC DC.SchDoc

File: PGS-System.PrjPcb

Date: 09/08/2024 | Page Num: 5

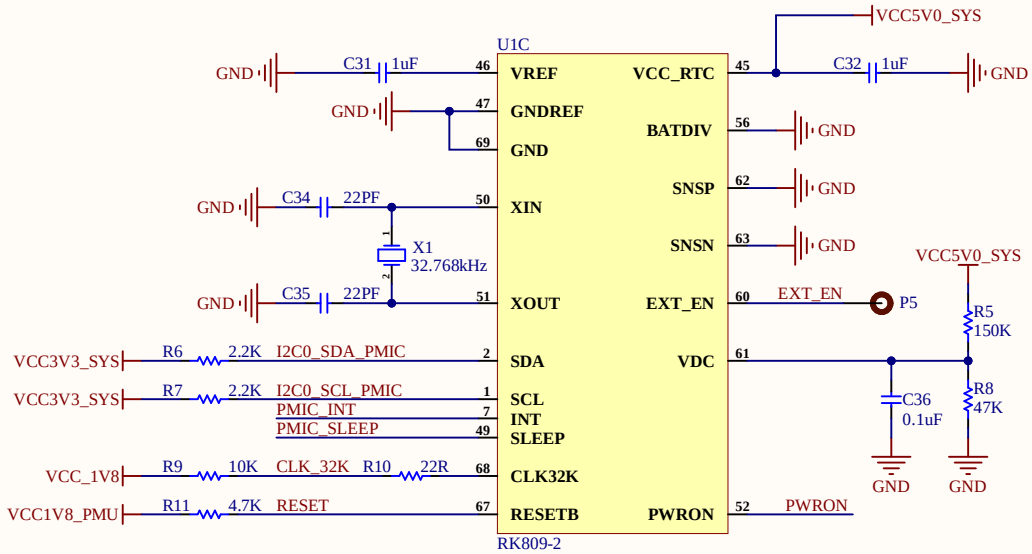
Drawn By: Huzaifa Ali | Page Total: 27

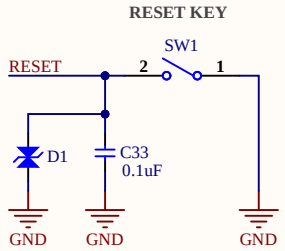
PMIC RK809 LDO

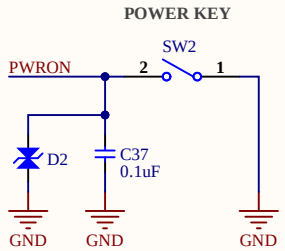


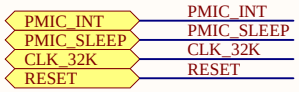
Title: POWER_PMIC_RK809_LDO.SchDoc	
File: PGS-System.PrjPcb	
Date: 09/08/2024	Page Num: 6
Drawn By: Huzaifa Ali	Page Total: 27

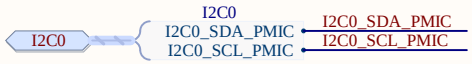
PMIC RK809 Management

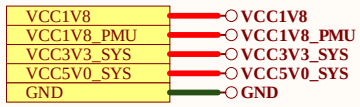








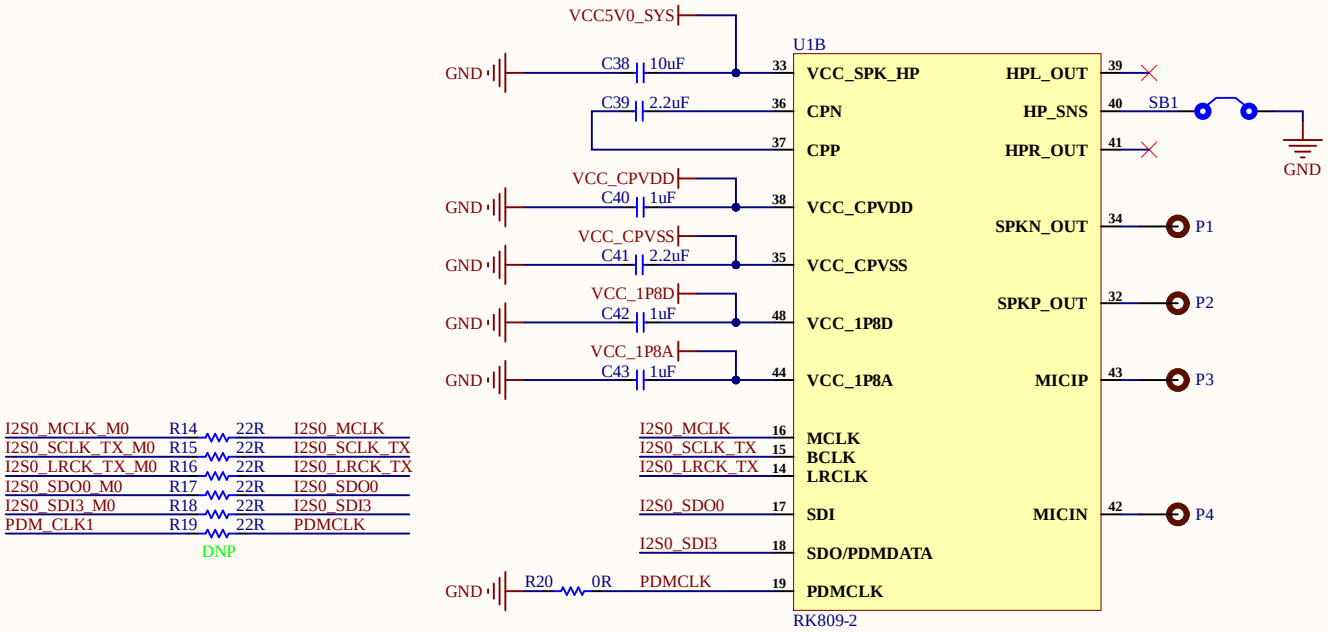


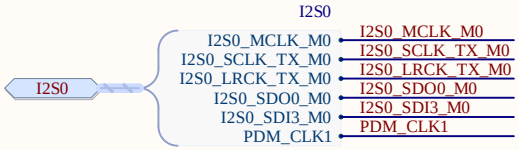


eSpark Consulting Group

Title: POWER_PMIC_RK809_MGM.SchDoc	
File: PGS-System.PrjPcb	
Date: 09/08/2024	Page Num: 7
Drawn By: Huzaifa Ali	Page Total: 27

PMIC RK809 CODEC



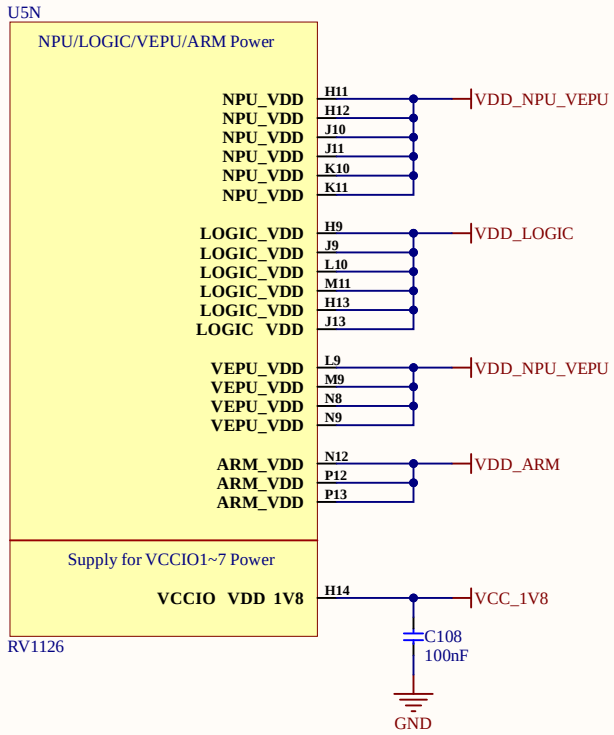


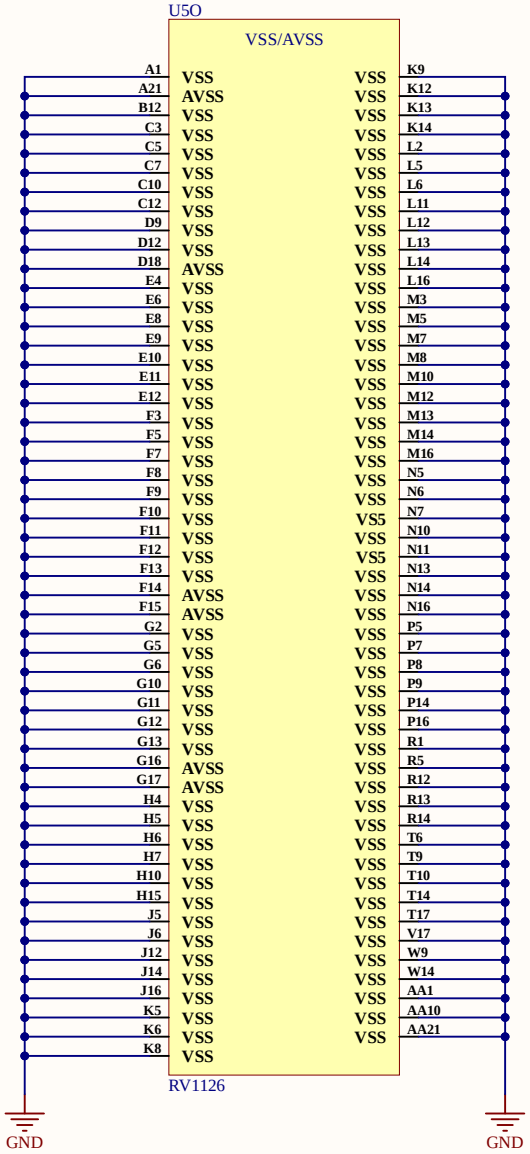


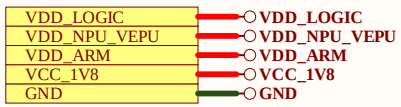
eSpark Consulting Group

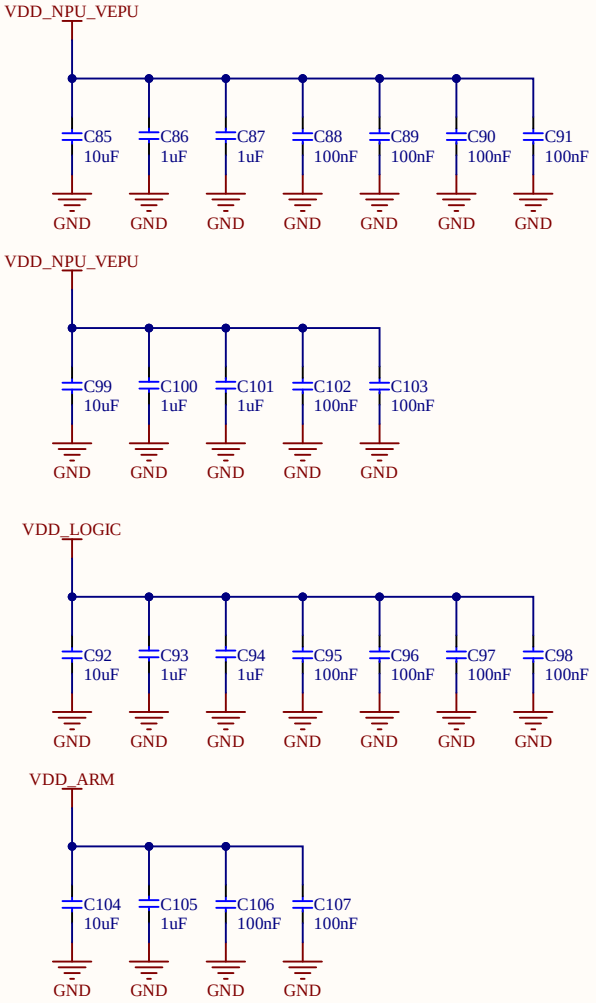
Title: POWER_PMIC_RK809_CODEC.SchDoc	
File: PGS-System.PrjPcb	
Date: 09/08/2024	Page Num: 8
Drawn By: Huzaifa Ali	Page Total: 27

RV1162-POWER AND GROUND









eSpark Consulting Group

Title: RV1126_POWER_SHEET.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 9
Drawn By: Huzaifa Ali	Page	Total: 27

RV1126_DRAM_CONTROLLER

U5A

DDR

DDR_D0	K4	DDR3/4_DQ0 LPDDR4_D8_A LPDDR3_D2
DDR_D1	J4	DDR3/4_DQ1 LPDDR4_D9_A LPDDR3_D6
DDR_D2	P4	DDR3/4_DQ2 LPDDR4_D10_A LPDDR3_D4
DDR_D3	N4	DDR3/4_DQ3 LPDDR4_D11_A LPDDR3_D0
DDR_D4	J3	DDR3/4_DQ4 LPDDR4_D12_A LPDDR3_D3
DDR_D5	K3	DDR3/4_DQ5 LPDDR4_D13_A LPDDR3_D7
DDR_D6	M4	DDR3/4_DQ6 LPDDR4_D14_A LPDDR3_D5
DDR_D7	P3	DDR3/4_DQ7 LPDDR4_D15_A LPDDR3_D1
DDR_DM0	N3	DDR3/4_DM0 LPDDR4_DM1_A LPDDR3_DM0
DDR_DQS0_P	L4	DDR3/4_DQS0P LPDDR4_DQS1P_A LPDDR3_DQS0P
DDR_DQS0_N	L3	DDR3/4_DQS0N LPDDR4_DQS1N_A LPDDR3_DQS0N
DDR_D8	D2	DDR3/4_DQ8 LPDDR4_D0_B LPDDR3_D10
DDR_D9	D1	DDR3/4_DQ9 LPDDR4_D1_B LPDDR3_D11
DDR_D10	H2	DDR3/4_DQ10 LPDDR4_D2_B LPDDR3_D8
DDR_D11	H1	DDR3/4_DQ11 LPDDR4_D3_B LPDDR3_D9
DDR_D12	G1	DDR3/4_DQ12 LPDDR4_D4_B LPDDR3_D12
DDR_D13	E1	DDR3/4_DQ13 LPDDR4_D5_B LPDDR3_D13
DDR_D14	C1	DDR3/4_DQ14 LPDDR4_D6_B LPDDR3_D14
DDR_D15	C2	DDR3/4_DQ15 LPDDR4_D7_B LPDDR3_D15
DDR_DM1	E2	DDR3/4_DM1 LPDDR4_DM0_B LPDDR3_DM1
DDR_DQS1_P	F2	DDR3/4_DQS1P LPDDR4_DQS0P_B LPDDR3_DQS1P
DDR_DQS1_N	F1	DDR3/4_DQS1N LPDDR4_DQS0N_B LPDDR3_DQS1N
DDR_D16	K1	DDR3/4_DQ16 LPDDR4_D0_A LPDDR3_D19
DDR_D17	K2	DDR3/4_DQ17 LPDDR4_D1_A LPDDR3_D18
DDR_D18	J1	DDR3/4_DQ18 LPDDR4_D2_A LPDDR3_D23
DDR_D19	J2	DDR3/4_DQ19 LPDDR4_D3_A LPDDR3_D22
DDR_D20	P2	DDR3/4_DQ20 LPDDR4_D4_A LPDDR3_D21
DDR_D21	L1	DDR3/4_DQ21 LPDDR4_D5_A LPDDR3_D16
DDR_D22	M1	DDR3/4_DQ22 LPDDR4_D6_A LPDDR3_D17
DDR_D23	P1	DDR3/4_DQ23 LPDDR4_D7_A LPDDR3_D20
DDR_DM2	M2	DDR3/4_DM2 LPDDR4_DM0_A LPDDR3_DM2
DDR_DQS2_P	N2	DDR3/4_DQS2P LPDDR4_DQS0P_A LPDDR3_DQS2P
DDR_DQS2_N	N1	DDR3/4_DQS2N LPDDR4_DQS0N_A LPDDR3_DQS2N
DDR_D24	A2	DDR3/4_DQ24 LPDDR4_D8_B LPDDR3_D30
DDR_D25	A3	DDR3/4_DQ25 LPDDR4_D9_B LPDDR3_D31
DDR_D26	G4	DDR3/4_DQ26 LPDDR4_D10_B LPDDR3_D28
DDR_D27	F4	DDR3/4_DQ27 LPDDR4_D11_B LPDDR3_D29
DDR_D28	H3	DDR3/4_DQ28 LPDDR4_D12_B LPDDR3_D24
DDR_D29	G3	DDR3/4_DQ29 LPDDR4_D13_B LPDDR3_D25
DDR_D30	B1	DDR3/4_DQ30 LPDDR4_D14_B LPDDR3_D26
DDR_D31	B2	DDR3/4_DQ31 LPDDR4_D15_B LPDDR3_D27
DDR_DM3	B3	DDR3/4_DM3 LPDDR4_DM1_B LPDDR3_DM3
DDR_DQS3_P	D3	DDR3/4_DQS3P LPDDR4_DQS1P_B LPDDR3_DQS3P
DDR_DQS3_N	E3	DDR3/4_DQS3N LPDDR4_DQS1N_B LPDDR3_DQS3N

RV1126

		LPDDR4_ODT1_B LPDDR3_A0 DDR4_ODT1 DDR3_A0
		LPDDR4_CKE0_A LPDDR3_A1 DDR4_A8 DDR3_A1
		LPDDR4_ODT1_A LPDDR3_A2 DDR4_BG1 DDR3_A2
		LPDDR4_CKE1_A LPDDR3_A3 DDR4_A0 DDR3_A3
		LPDDR4_ODT0_A LPDDR3_A4 DDR4_A7 DDR3_A4
		LPDDR4_A5_A LPDDR3_A5 DDR4_A9 DDR3_A5
		LPDDR4_CLKP_A LPDDR3_A6 DDR4_A4 DDR3_A6
		LPDDR4_A2_A LPDDR3_A7 DDR4_A11 DDR3_A7
		LPDDR4_CLKN_A LPDDR3_A8 DDR4_A10 DDR3_A8
		LPDDR4_A4_A LPDDR3_A9 DDR4_A2 DDR3_A9
		LPDDR4_A0_B DDR4_A15_CASn DDR3_A10
		LPDDR4_CS0n_B DDR4_A16_RASn DDR3_A11
		LPDDR4_ODT0_B DDR4_A5 DDR3_A12
		LPDDR4_A3_A DDR4_A13 DDR3_A13
		LPDDR4_CS1n_B DDR4_A14_Wen DDR3_A14
		LPDDR4_A5_B DDR4_A3 DDR3_A15
		DDR4_CS1n DDR3_BA0
		LPDDR4_CKE0_B DDR4_A1 DDR3_BA1
		LPDDR4_A0_A DDR4_A6 DDR3_BA2
		LPDDR4_A1_A DDR4_A12 DDR3_RASn
		LPDDR4_A1_B DDR4_CKE DDR3_CASn
		LPDDR4_CKE1_B DDR4_BA1 DDR3_WEn
		LPDDR4_CLKP_B LPDDR3_CLKP DDR4_CLKP DDR3_CLKP
		LPDDR4_CLKN_B LPDDR3_CLKN DDR4_CLKN DDR3_CLKN
		LPDDR4_A3_B LPDDR3_CKE DDR4_ACTn DDR3_CKE
		LPDDR4_A2_B LPDDR3_ODT0 DDR4_CS0n DDR3_ODT0
		LPDDR4_CS1n_A LPDDR3_ODT1 DDR4_BG0 DDR3_ODT1
		LPDDR4_A4_B LPDDR3_CS0n DDR4_ODT0 DDR3_CS0n
		LPDDR4_CS0n_A LPDDR3_CS1n DDR4_BA0 DDR3_CS1n
		DDR_RESETn
		DDR_RZQ
		DDR_AVSS
		DDR_VREF
		DDR_VDD
		DDR_VDD
		DDR_VDD
		DDR_VDD
		DDR_VDD
		DDR_VDD
		DDR_VDD

D10	DDR3_A0	
A8	DDR3_A1	
D11	DDR3_A2	
C9	DDR3_A3	
B8	DDR3_A4	
B9	DDR3_A5	
A9	DDR3_A6	
C11	DDR3_A7	
A10	DDR3_A8	
B11	DDR3_A9	
B5	DDR3_A10	
B10	DDR3_A11	
A7	DDR3_A12	
A12	DDR3_A13	
A11	DDR3_A14	
B6	DDR3_A15	
D8	DDR3_BA0	
B7	DDR3_BA1	
C8	DDR3_BA2	
D7	DDR3_RASN	
C6	DDR3_CASN	
A6	DDR3_WEN	
D4	DDR3_CLK_P	
D5	DDR3_CLK_N	
A4	DDR3_CKE	
B4	DDR3_ODT0	
D6	DDR3_ODT1	
A5	DDR3_CS0N	
E7	DDR3_CS1N	
C4	DDR3_RST	
E5	R35 240R	GND
F6		GND
M6	VREF_DDR	
G7	VCC_DDR	
G8		
G9		
H8		
J7		
K7		
L7		

VCC_DDR

C109 10uF	C110 1uF	C111 1uF	C112 1uF	C113 100nF	C114 100nF	C115 100nF	C159 100nF
GND	GND	GND	GND	GND	GND	GND	GND

DDR3

DDR_D[31..0]	D[31..0]
DDR3_A[15..0]	A[15..0]
DDR_DQS0_P	DQS0_P
DDR_DQS0_N	DQS0_N
DDR_DQS1_P	DQS1_P
DDR_DQS1_N	DQS1_N
DDR_DQS2_P	DQS2_P
DDR_DQS2_N	DQS2_N
DDR_DQS3_P	DQS3_P
DDR_DQS3_N	DQS3_N
DDR_DM[3..0]	DM[3..0]
DDR3_BA[2..0]	BA[2..0]
DDR3_RASN	RASN
DDR3_CASN	CASN
DDR3_WEN	WEN
DDR3_RST	RST
DDR3_CKE	CKE
DDR3_ODT0	ODT0
DDR3_ODT1	ODT1
DDR3_CS0N	CS0N
DDR3_CS1N	CS1N
DDR3_CLK_P	CLK_P
DDR3_CLK_N	CLK_N

DDR3

VREF_DDR	VREF_DDR
VCC_DDR	VCC_DDR
GND	GND

eSpark Consulting Group

Title: RV1126_DRAM_SHEET.SchDoc	
File: PGS-System.PrjPcb	
Date: 09/08/2024	Page Num: 10
Drawn By: Huzaifa Ali	Page Total: 27

RV1126_SARADC_UART-DEBUG_SHEET

U5E

LCDC/RGMII/CIF/UART/JTAG/I2S/SPI/I2C

UART4_RTSN_M1 RGMII_TXD3_M1 CIF_D0_M1 LCDC_D0 GPIO2_A4_d	J18	ICR_AIN1
I2C5_SCL_M0 UART4_CTSN_M1 RGMII_CRS_M1 CIF_D1_M1 LCDC_D1 GPIO2_A5_d	L17	ICR_AIN2
UART4_TX_M1 PWM5_M1 RGMII_COL_M1 CIF_D2_M1 LCDC_D2 GPIO2_A6_d	M21	
I2S2_SDO_M1 UART4_RX_M1 PWM4_M1 SPI0_CS0n_M2 LCDC_D3 GPIO2_A7_d	M20	
I2S2_SDI_M1 UART5_TX_M1 PWM3_IR_M1 SPI0_MOSI_M2 LCDC_D4 GPIO2_B0_d	L19	
I2S2_SCLK_M1 UART5_RX_M1 PWM2_M1 SPI0_MISO_M2 LCDC_D5 GPIO2_B1_d	L20	
I2S2_LRCK_M1 UART5_RTSN_M1 PWM1_M1 SPI0_CLK_M2 LCDC_D6 GPIO2_B2_d	K16	
I2C5_SDA_M0 I2S2_MCLK_M1 UART5_CTSN_M1 PWM0_M1 SPI0_CS1n_M2 CIF_D3_M1 LCDC_D7 GPIO2_B3_d	K17	
RGMII_RXDV_M1 CIF_D4_M1 LCDC_D8 GPIO2_B4_d	K18	
RGMII_RXD0_M1 CIF_D5_M1 LCDC_D9 GPIO2_B5_d	K19	
RGMII_RXD1_M1 CIF_D6_M1 LCDC_D10 GPIO2_B6_d	K20	
RGMII_CLK_M1 CIF_D7_M1 LCDC_D11 GPIO2_B7_d	K21	
RGMII_RXER_M1 CIF_D8_M1 LCDC_D12 GPIO2_C0_d	J19	
RGMII_MDIO_M1 CIF_D9_M1 LCDC_D13 GPIO2_C1_d	J21	
RGMII_MDC_M1 CIF_D10_M1 LCDC_D14 GPIO2_C2_d	J20	
RGMII_TXD0_M1 CIF_D11_M1 LCDC_D15 GPIO2_C3_d	H20	
RGMII_TXD1_M1 CIF_D12_M1 LCDC_D16 GPIO2_C4_d	H19	
CLK_OUT_ETHERNET_M1 CIF_D13_M1 LCDC_D17 GPIO2_C5_d	G21	
RGMII_TXEN_M1 CIF_D14_M1 LCDC_D18 GPIO2_C6_d	G20	
I2S1_MCLK_M2 RGMII_RXD2_M1 CIF_D15_M1 LCDC_D19 GPIO2_C7_d	G19	ICR_BIN2
I2S1_SDO_M2 RGMII_RXD3_M1 CIF_VSYNC_M1 LCDC_D20 GPIO2_D0_d	H18	ICR_BIN1
I2S1_SCLK_M2 RGMII_TXD2_M1 CIF_CLKOUT_M1 LCDC_D21 GPIO2_D1_d	F21	W0RK_LED
I2S1_LRCK_M2 RGMII_TXCLK_M1 CIF_CLKIN_M1 LCDC_D22 GPIO2_D2_d	F20	
I2S1_SDI_M2 RGMII_RXCLK_M1 CIF_HSYNC_M1 LCDC_D23 GPIO2_D3_d	F19	
I2C3_SCL_M1 PWM6_M1 SPI1_CS0n_M2 LCDC_DEN GPIO2_D4_d	J17	
I2C3_SDA_M1 PWM10_M1 SPI1_CLK_M2 LCDC_HSYNC GPIO2_D5_d	H17	
UART3_RTSN_M2 PWM9_M1 SPI1_MOSI_M2 LCDC_VSYNC GPIO2_D6_d	C21	
UART3_CTSN_M2 PWM8_M1 SPI1_MISO_M2 LCDC_CLK GPIO2_D7_d	D21	
I2C4_SCL_M0 CAN_RXD_M0 UART3_TX_M2 PWM7_IR_M1 SPI1_CS1n_M2 GPIO3_A0_u	E20	
I2C4_SDA_M0 CAN_TXD_M0 UART3_RX_M2 PWM11_IR_M1 GPIO3_A1_u	E19	
A7_JTAG_TCK_M1 UART2_TX_M1 GPIO3_A2_u	G18	UART2_RX_M0_DEBUG
A7_JTAG_TMS_M1 UART2_RX_M1 GPIO3_A3_u	H16	UART2_TX_M0_DEBUG
VCCIO5_VDD1	J15	
VCCIO5_VDD1	K15	

RV1126

VCC_3V3

C205 100nF

C206 100nF

GND GND

U5C

ADC

ADCIN0	E17	ADKEY_IN0
ADCIN1	D17	BOM_ID
ADCIN2	B18	LIGHT_SENSOR1
ADCIN3	A18	LIGHT_SENSOR2
ADCIN4	C17	
ADCIN5	B17	
ADC_AVDD_1V8	E16	VCC_1V8

RV1126

C117 100nF

GND

UART2_RX_M0_DEBUG — UART2_RX_M0_DEBUG

UART2_TX_M0_DEBUG — UART2_TX_M0_DEBUG

LIGHT_SENSOR1 — LIGHT_SENSOR1

LIGHT_SENSOR2 — LIGHT_SENSOR2

ICR_AIN1 — ICR_AIN1

ICR_BIN1 — ICR_BIN1

ICR_AIN2 — ICR_AIN2

ICR_BIN2 — ICR_BIN2

ESC/RECOVERY

VCC_1V8

R36 10K

ADKEY_IN0 — R37 100R — SW3 (1, 2, 3, 4) TSA016A2518A

C116 1nF

D3 ESD5451N

GND GND GND

WORK_LED

W0RK_LED

R121 510R

LED2 Red

GND

BOM_ID

VCC_1V8

R38 10K DNP

BOM_ID

C118 1nF DNP

R39 10K DNP

GND GND

VCC3V3 — VCC3V3

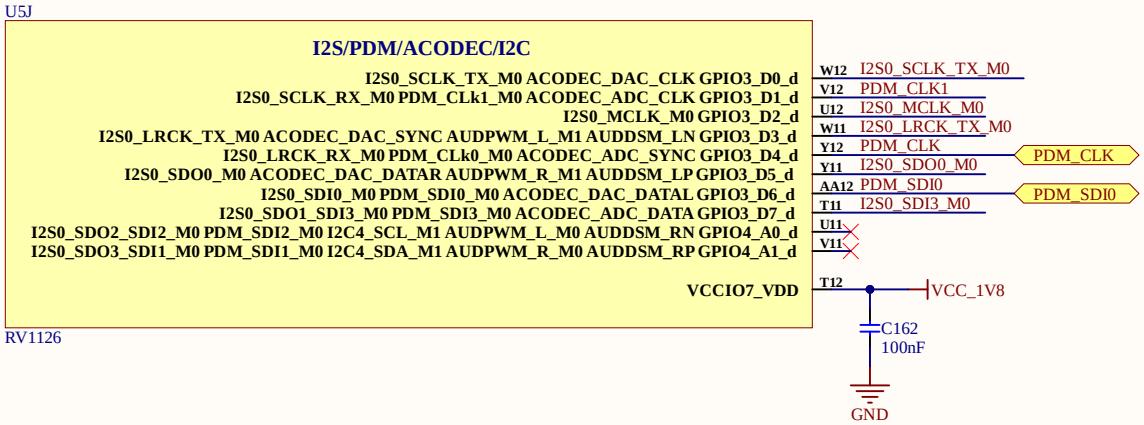
VCC1V8 — VCC1V8

GND — GND

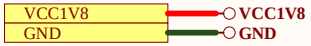
eSpark Consulting Group

Title: RV1126_SARADC_SHEET.SchDoc	
File: PGS-System.PrjPcb	
Date: 09/08/2024	Page Num: 11
Drawn By: Huzaifa Ali	Page Total: 27

RV1126_AUDIO_SHEET



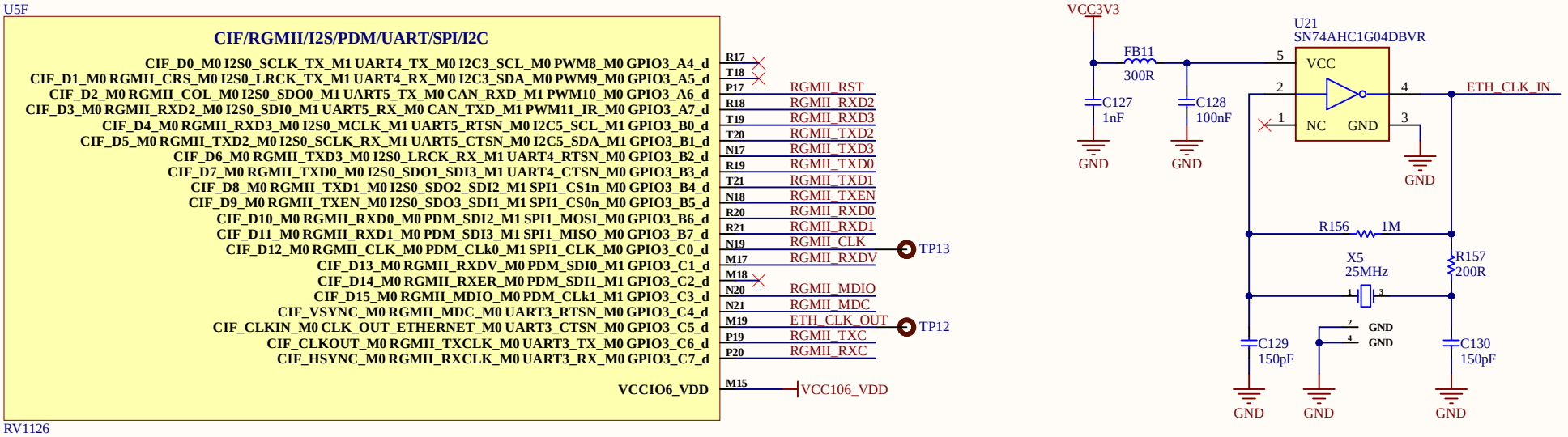




eSpark Consulting Group

Title: RV1126_AUDIO_SHEET.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page Num:	12
Drawn By: Huzaifa Ali	Page Total:	27

RV1162_ETHERNET_SHEET



Title:	RV1126_ETHERNET_SHEET.SchDoc
File:	PGS-System.PrjPcb
Date: 09/08/2024	Page Num: 13
Drawn By: Huzaifa Ali	Page Total: 27

RV1162-eMMC_SPI-FLASH

Flash Power Manage

U5L

I2S1/FSPI/EMMC/FLASH

EMMC_D0 FLASH_D0 GPIO0_C4_u	W2	eMMC_DATA0
EMMC_D1 FLASH_D1 GPIO0_C5_u	W1	eMMC_DATA1
EMMC_D2 FLASH_D2 GPIO0_C6_u	V4	eMMC_DATA2
EMMC_D3 FLASH_D3 GPIO0_C7_u	V3	eMMC_DATA3
EMMC_D4 FLASH_D4 GPIO0_D0_u	V2	eMMC_DATA4
FSPI_CS1n EMMC_D5 FLASH_D5 GPIO0_D1_u	V1	eMMC_DATA5
EMMC_D6 FLASH_D6 GPIO0_D2_u	U4	eMMC_DATA6
EMMC_D7 FLASH_D7 GPIO0_D3_u	U3	eMMC_DATA7
I2S1_MCLK_M0 FSPI_CS0n FLASH_CS0n GPIO0_D4_u	U2	FSPI_CS0n
EMMC_CMD FLASH_WRn GPIO0_D5_u	T4	eMMC_CMD
I2S1_SDO_M0 FSPI_D2 GPIO0_D6_d	T3	FSPI_D2
EMMC_CLKO FLASH_CLE GPIO0_D7_d	T1	eMMC_CLK
I2S1_LRCK_M0 FSPI_D0 FLASH_ALE GPIO1_A0_d	T2	FSPI_D0
I2S1_SCLK_M0 FSPI_D1 FLASH_RDYn GPIO1_A1_u	R2	FSPI_D1
I2S1_SDI_M0 FSPI_D3 FLASH_RDn GPIO1_A2_u	R4	FSPI_D3
FSPI_CLK EMMC_RSTn FLASH_WPn GPIO1_A3_d	R3	eMMC_RSTN

R24 0R FSPI_CLK

VCCIO1_VDD — P6 — VCCIO_FLASH

C164 100nF GND

RV1126

VCC_3V3

R91 4.7K

FLASH_VOL_SEL

R92 10K DNP

GND

NOTE:
FLASH(VCCIO1) power domain IO supply configuration pin:

Condition	VCCIO1 (VCCIO_FLASH)
FLASH_VOL_SEL=0	3.3V
FLASH_VOL_SEL=1	1.8V Default

VCCIO_FLASH

C163 4.7uF

R82 0R VCC_1V8

R89 0R VCC_3V3 DNP

GND

JTAG

RISC-V_JTAG_TRSTn	P6
RISC-V_JTAG_TCK	P7
RISC-V_JTAG_TMS	P8
RISC-V_JTAG_TDO	P9
RISC-V_JTAG_TDI	P10
GND	P11

U5I

SDMMC0/UART/JTAG

UART2_RX_M0 TEST_CLK1_OUT SDMMC0_D0 GPIO1_A4_u	Y14	
UART2_TX_M0 TEST_CLK0_OUT MCU_JTAG_TRSTn SDMMC0_D1 GPIO1_A5_u	W13	RISC-V_JTAG_TRSTn
UART3_RX_M1 A7_JTAG_TCK_M0 MCU_JTAG_TCK SDMMC0_D2 GPIO1_A6_u	V13	RISC-V_JTAG_TCK
UART3_TX_M1 A7_JTAG_TMS_M0 MCU_JTAG_TMS SDMMC0_D3 GPIO1_A7_u	U13	RISC-V_JTAG_TMS
UART3_RTSN_M1 MCU_JTAG_TDO SDMMC0_CLK GPIO1_B0_u	AA13	RISC-V_JTAG_TDO
UART3_CTSN_M1 MCU_JTAG_TDI SDMMC0_CMD GPIO1_B1_u	Y13	RISC-V_JTAG_TDI

VCCIO2_VDD — T13 — VCCIO_SD

C54 100nF GND

RV1126

VCC1V8	VCC1V8
VCC_3V3	VCC_3V3
VCCIO_FLASH	VCCIO_FLASH
VCCIO_SD	VCCIO_SD
GND	GND

eMMC

eMMC_DATA[7..0]	DATA[7..0]
eMMC_CLK	CLK
eMMC_CMD	CMD
eMMC_RSTN	RST

→ eMMC

FSPI

FSPI_CLK	FSPI_CLK
FSPI_D0	FSPI_D0
FSPI_D1	FSPI_D1
FSPI_D2	FSPI_D2
FSPI_D3	FSPI_D3
FSPI_CS0n	FSPI_CS0n

→ FSPI

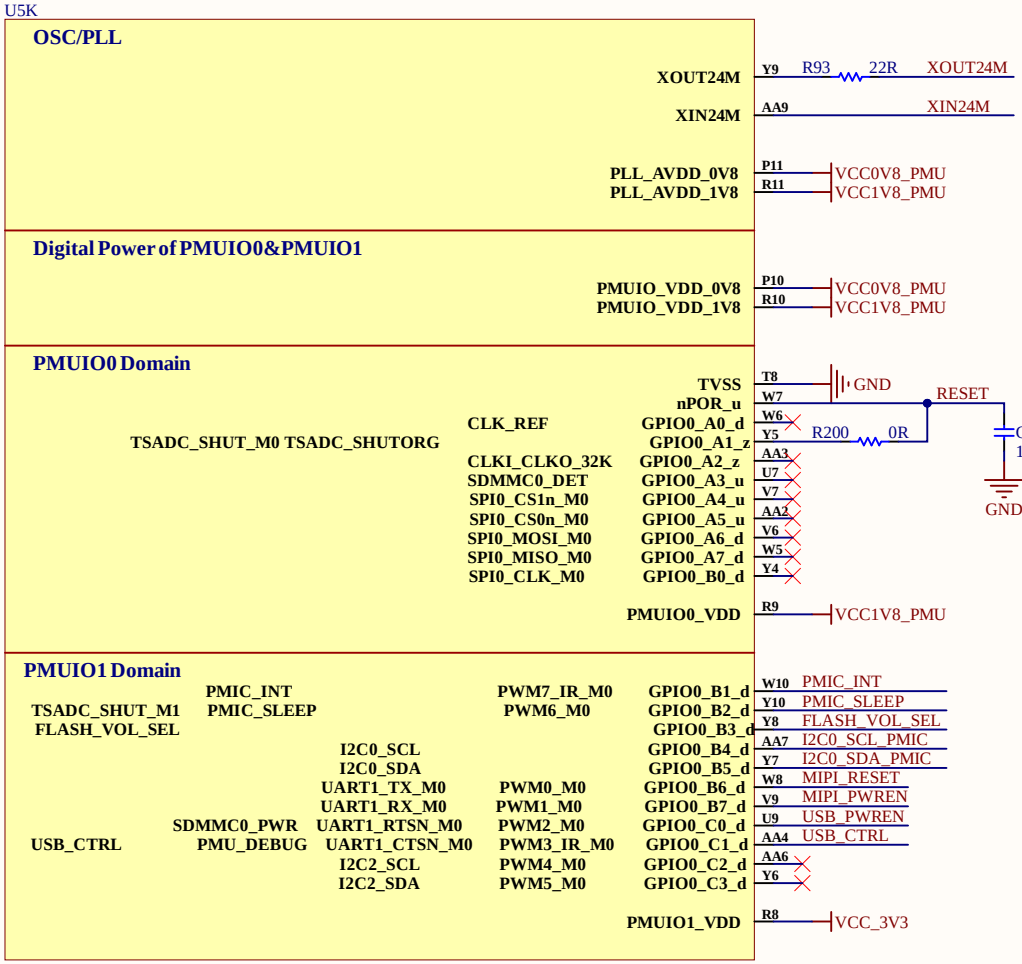
eSpark Consulting Group

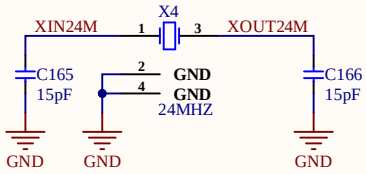
Title: RV1126_eMMC_SPI-FLASH.SchDoc

File: PGS-System.PrjPcb

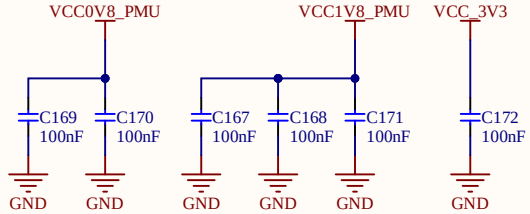
Date: 09/08/2024	Page Num: 14
Drawn By: Huzaifa Ali	Page Total: 27

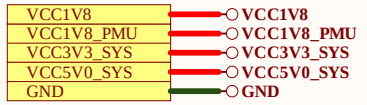
RV1126_OSC/PLL/PMUIO

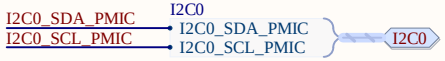




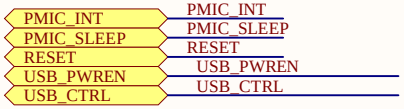
Note:
Decoupling Capacitor place close to the pin

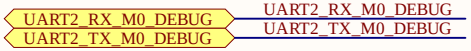








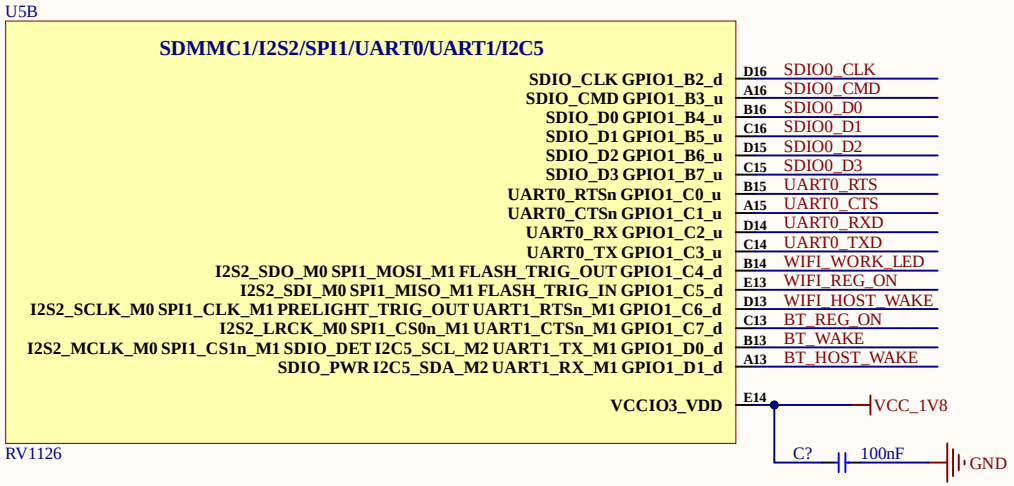


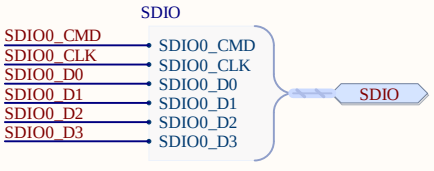


eSpark Consulting Group

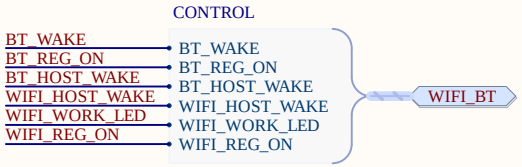
Title: RV1126_OSC-PLL-PMUIO.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 15
Drawn By: Huzaifa Ali	Page	Total: 27

RV1162_SDMMC SHEET











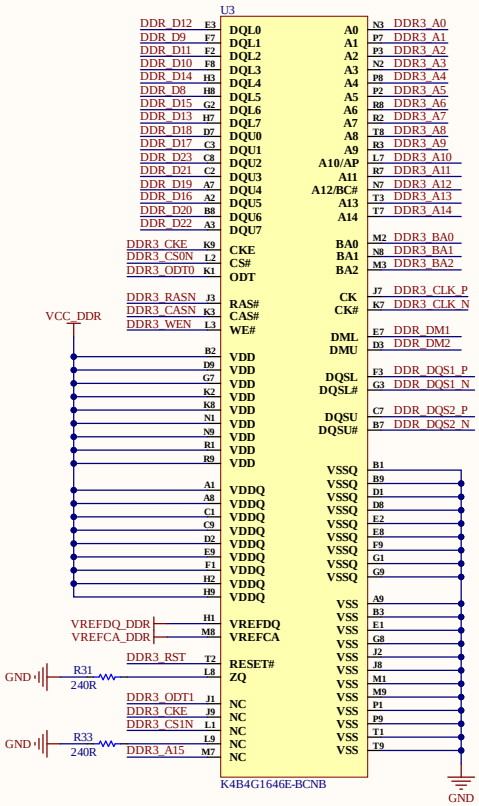
eSpark Consulting Group

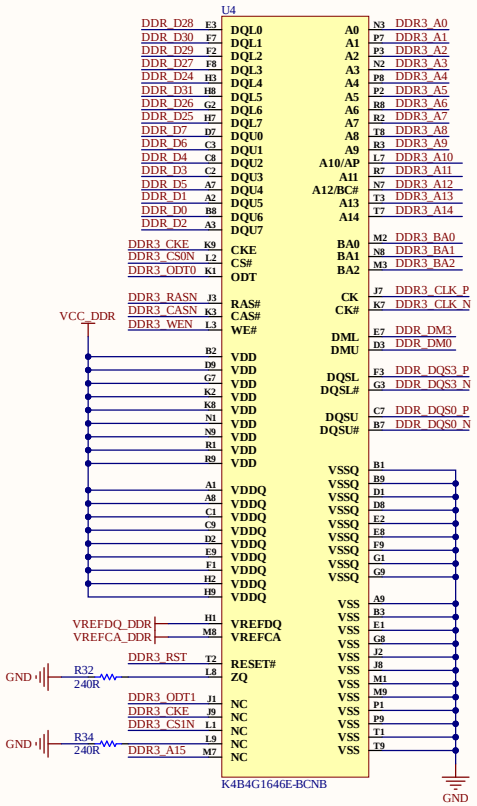
Title: RV1126_SDMMC_SHEET.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 16
Drawn By: Huzaifa Ali	Page	Total: 27

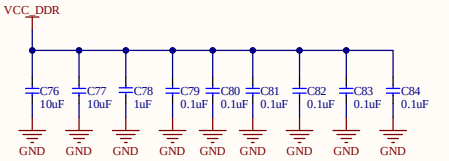
DDR3_SDRAM INTERFACE

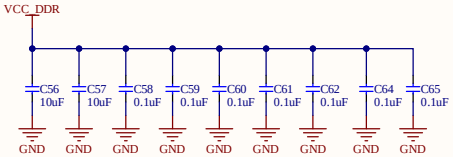
DDR3 2x16bit

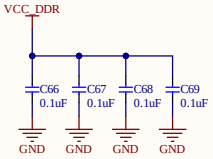
NOTE:
All the Power filter capacitors should be placed close to the power pins of DDR3

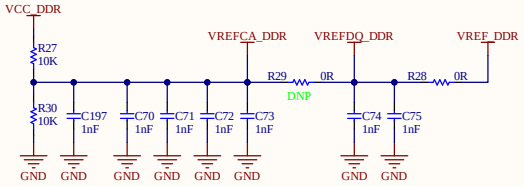


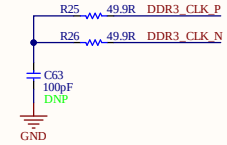


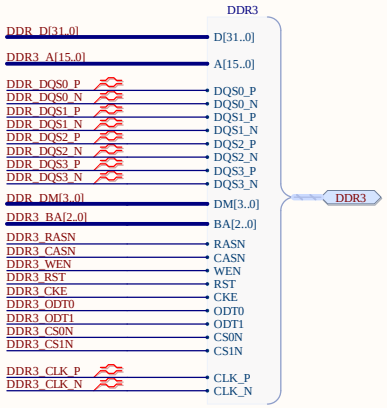


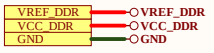








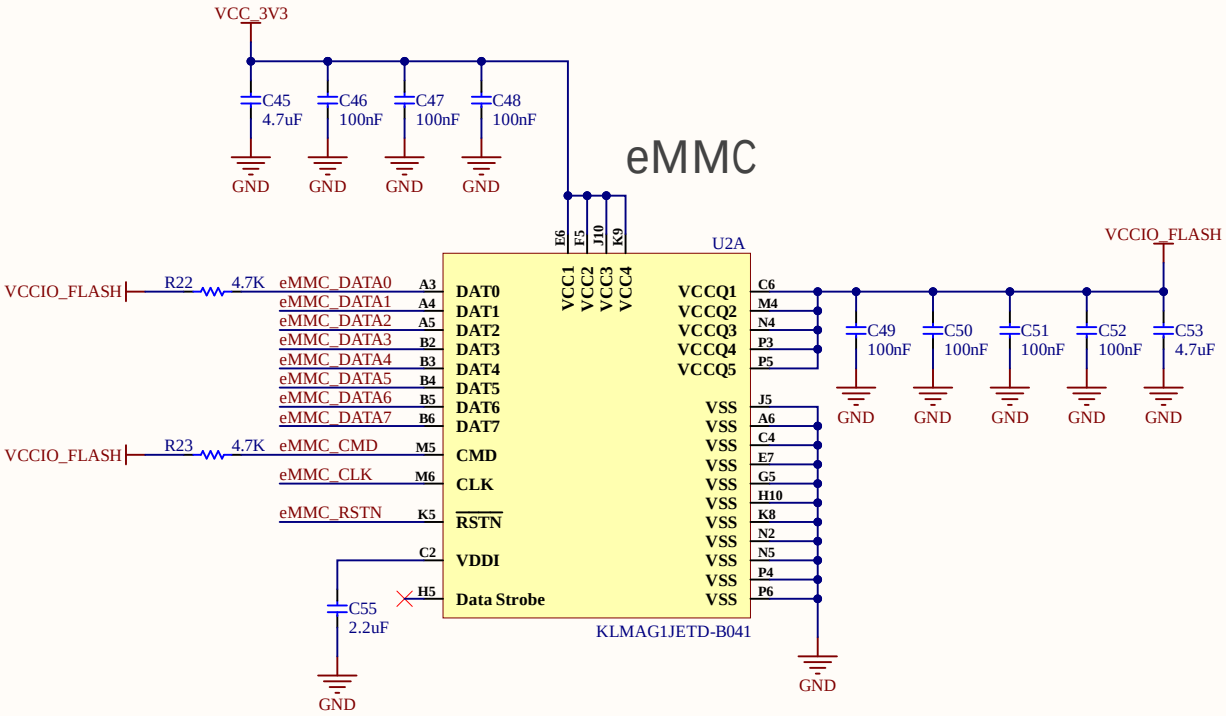


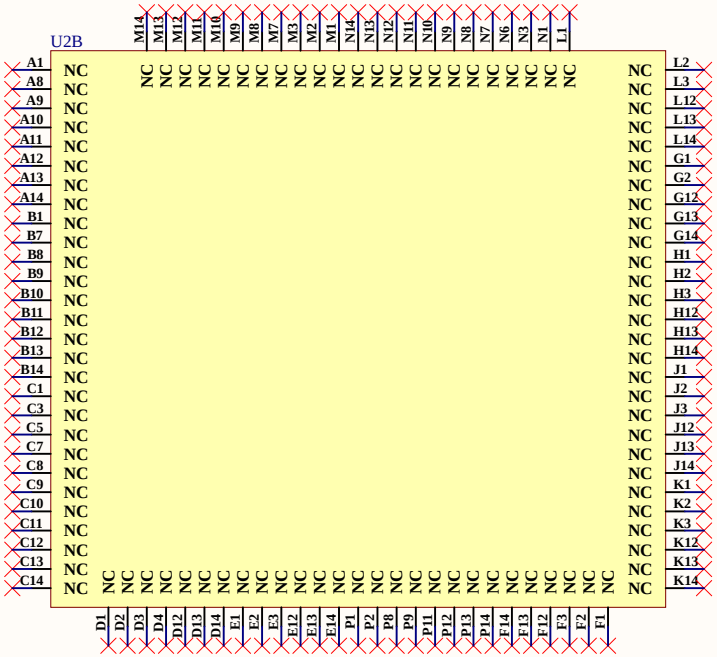


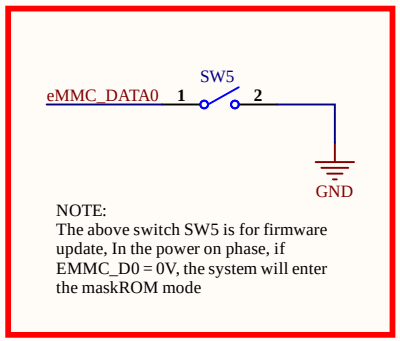
eSpark Consulting Group

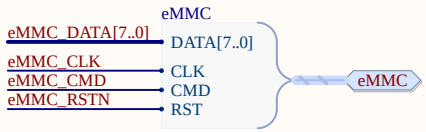
Title: DDR3_SDRAM_SHEET.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 17
Drawn By: Huzaifa Ali	Page	Total: 27

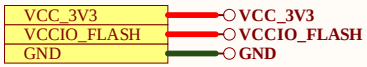
eMMC_INTERFACE







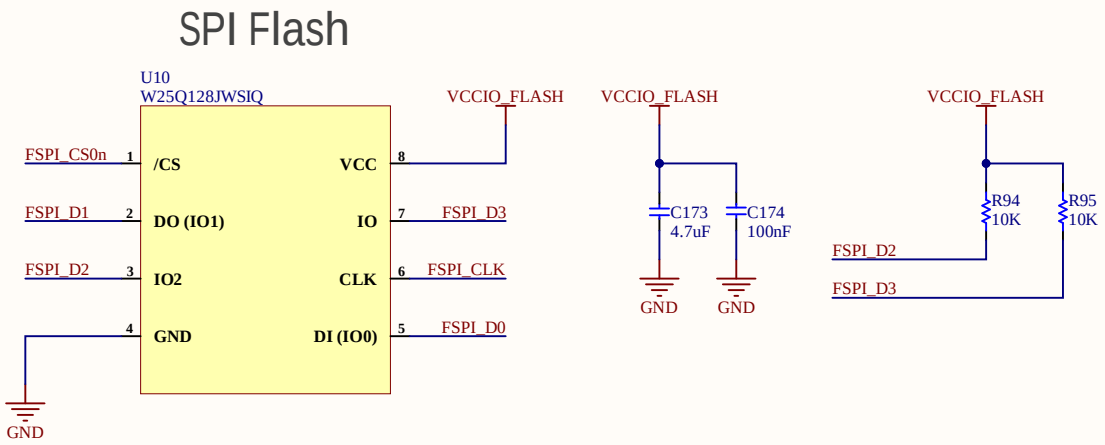


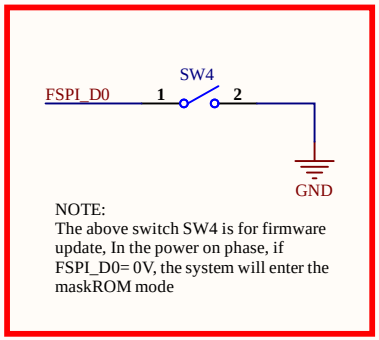


eSpark Consulting Group

Title: eMMC_INTERFACE.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page Num:	18
Drawn By: Huzaifa Ali	Page Total:	27

SPI_FLASH_INTERFACE





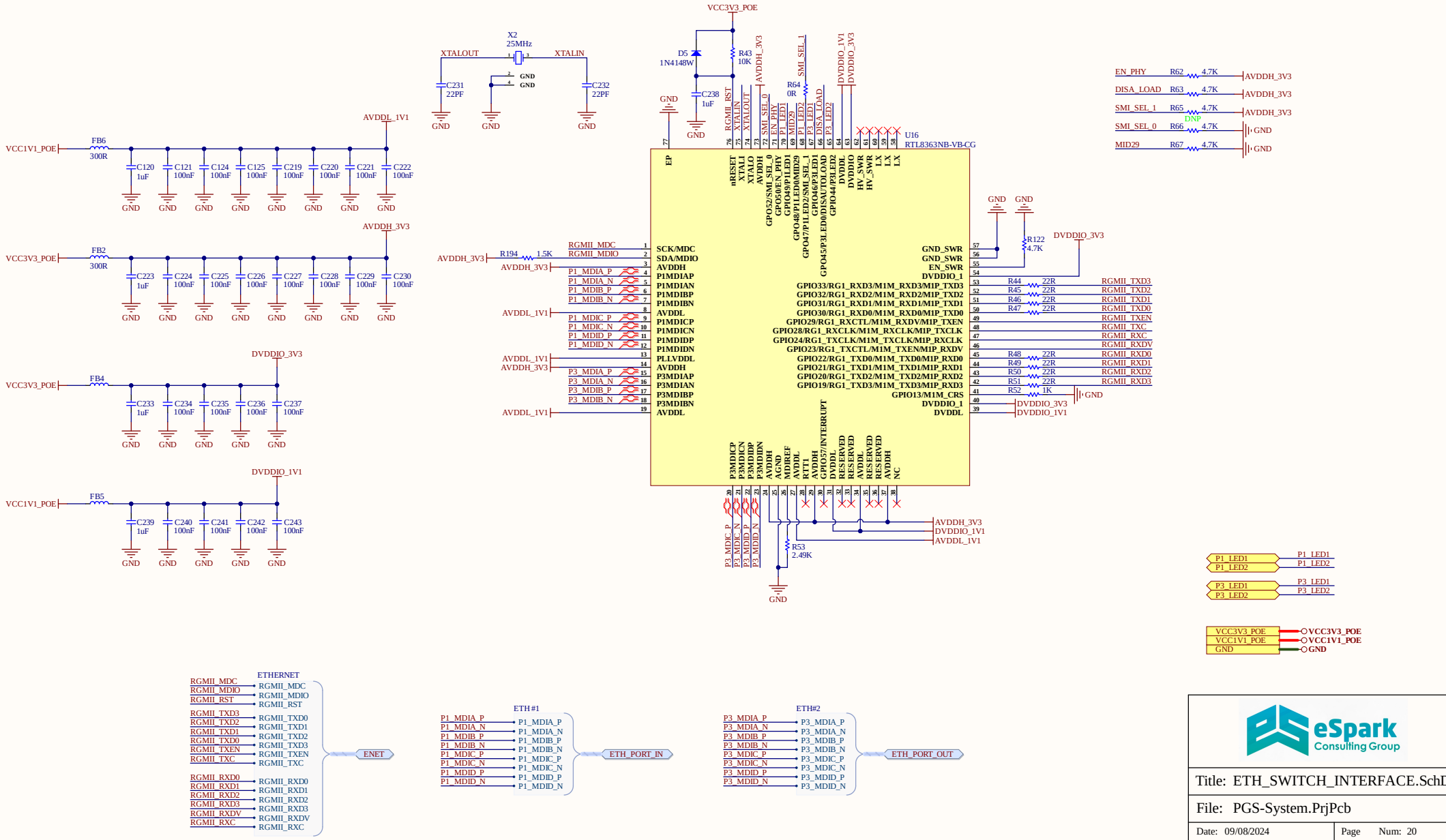




eSpark Consulting Group

Title: SPI_FLASH_INTERFACE.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 19
Drawn By: Huzaifa Ali	Page	Total: 27

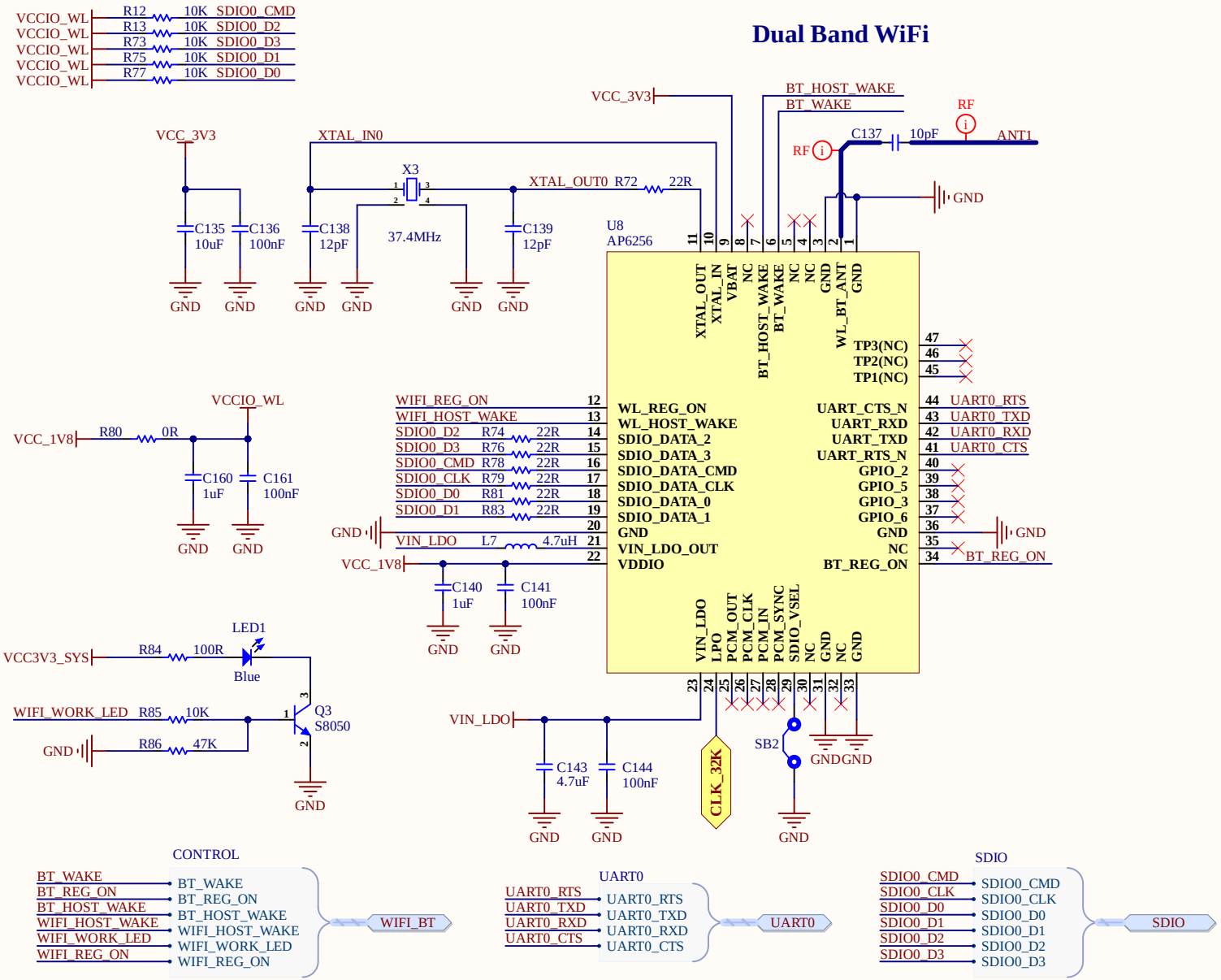
ETHERNET_SWITCH_INTERFACE



Title:	ETH_SWITCH_INTERFACE.SchDoc
File:	PGS-System.PrjPcb
Date: 09/08/2024	Page Num: 20
Drawn By: Huzaifa Ali	Page Total: 27

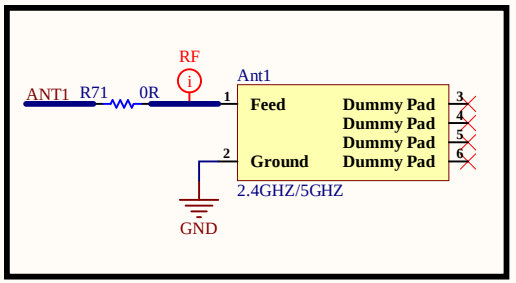
WIFI_BLUETOOTH_INTERFACE

Dual Band WiFi

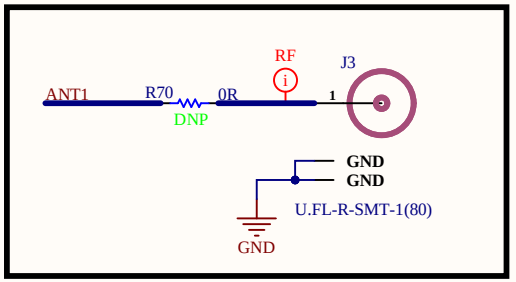


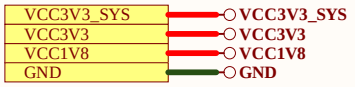
WIFI ANTENNA

OPTION#1



OPTION#2





eSpark Consulting Group

Title: WIFI_AP6256_SHEET.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 21
Drawn By: Huzaifa Ali	Page	Total: 27

MIPI-CSI_CAMERA_INTERFACE

U5H

MIPI CSI RX1

MIPI_CSI_RX1_D0P LVDS1_RX0P	AA20	MIPI_CSI_RX1_D0P
MIPI_CSI_RX1_D0N LVDS1_RX0N	Y20	MIPI_CSI_RX1_D0N
MIPI_CSI_RX1_D1P LVDS1_RX1P	AA19	MIPI_CSI_RX1_D1P
MIPI_CSI_RX1_D1N LVDS1_RX1N	Y19	MIPI_CSI_RX1_D1N
MIPI_CSI_RX1_D2P LVDS1_RX2P	AA18	MIPI_CSI_RX1_D2P
MIPI_CSI_RX1_D2N LVDS1_RX2N	Y18	MIPI_CSI_RX1_D2N
MIPI_CSI_RX1_D3P LVDS1_RX3P	Y17	MIPI_CSI_RX1_D3P
MIPI_CSI_RX1_D3N LVDS1_RX3N	W17	MIPI_CSI_RX1_D3N
MIPI_CSI_RX1_CLKP LVDS1_CLKP	V18	MIPI_CSI_RX1_CLKP
MIPI_CSI_RX1_CLKN LVDS1_CLKN	W18	MIPI_CSI_RX1_CLKN
MIPI_CSI_RX1_AVDD_0V8	R15	MIPI_CSI_AVDD_0V8
MIPI_CSI_RX1_AVDD_1V8	R16	MIPI_CSI_AVDD_1V8

MIPI CSI RX0

MIPI_CSI_RX0_D0P LVDS0_RX0P	V16	MIPI_CSI_RX0_D0P
MIPI_CSI_RX0_D0N LVDS0_RX0N	U16	MIPI_CSI_RX0_D0N
MIPI_CSI_RX0_D1P LVDS0_RX1P	Y16	MIPI_CSI_RX0_D1P
MIPI_CSI_RX0_D1N LVDS0_RX1N	W16	MIPI_CSI_RX0_D1N
MIPI_CSI_RX0_D2P LVDS0_RX2P	W15	MIPI_CSI_RX0_D2P
MIPI_CSI_RX0_D2N LVDS0_RX2N	Y15	MIPI_CSI_RX0_D2N
MIPI_CSI_RX0_D3P LVDS0_RX3P	AA15	MIPI_CSI_RX0_D3P
MIPI_CSI_RX0_D3N LVDS0_RX3N	AA16	MIPI_CSI_RX0_D3N
MIPI_CSI_RX0_CLKP LVDS0_CLKP	U15	MIPI_CSI_RX0_CLKP
MIPI_CSI_RX0_CLKN LVDS0_CLKN	V15	MIPI_CSI_RX0_CLKN
MIPI_CSI_RX0_AVDD_0V8	T15	MIPI_CSI_AVDD_0V8
MIPI_CSI_RX0_AVDD_1V8	T16	MIPI_CSI_AVDD_1V8

RV1126

U5G

SPI/I2C/I2S/UART/MIPI CLK

I2C1_SDA UART4_RTSN_M2 GPIO1_D2_u	W19	I2C1_SDA	R203	2.2K	VCCI04_VDD
I2C1_SCL UART4_CTSN_M2 GPIO1_D3_u	Y21	I2C1_SCL	R204	2.2K	VCCI04_VDD
UART4_RX_M2 GPIO1_D4_d	W20	MIPI_CSI_PWDN0			
SPI0_CS1n_M1 I2S1_MCLK_M1 UART4_TX_M2 GPIO1_D5_d	V20	MIPI_CSI_PWDN1			
SPI0_MOSI_M1 I2S1_SCLK_M1 I2C3_SCL_M2 GPIO1_D6_d	V19	I2C3_SDA	R205	2.2K	VCCI04_VDD
SPI0_MISO_M1 I2S1_LRCK_M1 I2C3_SDA_M2 GPIO1_D7_d	U18	I2C3_SCL	R206	2.2K	VCCI04_VDD
SPI0_CS0n_M1 I2S1_SDI_M1 UART5_TX_M2 GPIO2_A0_d	U19	MIPI_CSI_RST0			
SPI0_CLK_M1 I2S1_SDO_M1 UART5_RX_M2 GPIO2_A1_d	U20	MIPI_CSI_RST1			
MIPI_CSI_CLK1 UART5_RTSN_M2 GPIO2_A2_d	W21	MIPI_CSI_CLK1			
MIPI_CSI_CLK0 UART5_CTSN_M2 GPIO2_A3_d	V21	MIPI_CSI_CLK0			
VCCIO4_VDD	P15	VCCI04_VDD			

RV1126

C212 100nF

GND

MIPI-CSI_CAMERA #1

J4

GND	1	VCC2V8_AVDD	C145 10uF, C146 100nF
MIPI_CSI_PWDN1	2		
MIPI_CSI_RST1	3		
I2C1_SDA	4		
I2C1_SCL	5		
GND	6		
MIPI_CSI_RX1_D0N	7		
MIPI_CSI_RX1_D0P	8		
MIPI_CSI_RX1_D1N	9		
MIPI_CSI_RX1_D1P	10		
MIPI_CSI_RX1_D2N	11		
MIPI_CSI_RX1_D2P	12		
MIPI_CSI_RX1_D3N	13		
MIPI_CSI_RX1_D3P	14		
GND	15		
MIPI_CSI_RX1_CLKN	16	VCC1V2_DVDD	C147 10uF, C148 100nF
MIPI_CSI_RX1_CLKP	17		
GND	18		
MIPI_CSI_CLK1	19		
GND	20		
VCC1V8_DOVDD	21	VCC1V8_DOVDD	C149 100nF
VCC1V2_DVDD	22		
VCC2V8_AVDD	23		
	24		
	MP1		
	MP2		

FH12-24S-0.5SH(55)

MIPI-CSI_CAMERA #2

J5

GND	1	VCC2V8_AVDD	C150 10uF, C151 100nF
MIPI_CSI_PWDN0	2		
MIPI_CSI_RST0	3		
I2C3_SDA	4		
I2C3_SCL	5		
GND	6		
MIPI_CSI_RX0_D0N	7		
MIPI_CSI_RX0_D0P	8		
MIPI_CSI_RX0_D1N	9		
MIPI_CSI_RX0_D1P	10		
MIPI_CSI_RX0_D2N	11		
MIPI_CSI_RX0_D2P	12		
MIPI_CSI_RX0_D3N	13		
MIPI_CSI_RX0_D3P	14		
GND	15		
MIPI_CSI_RX0_CLKN	16	VCC1V2_DVDD	C152 10uF, C153 100nF
MIPI_CSI_RX0_CLKP	17		
GND	18		
MIPI_CSI_CLK0	19		
GND	20		
VCC1V8_DOVDD	21	VCC1V8_DOVDD	C154 100nF
VCC1V2_DVDD	22		
VCC2V8_AVDD	23		
	24		
	MP1		
	MP2		

FH12-24S-0.5SH(55)

MIPI_CSI_AVDD_1V8 — R88 0R — VCC_1V8; C157 100nF, C158 100nF

VCC_1V8 — R109 0R — VCCI04_VDD; C132 100nF

MIPI_CSI_AVDD_1V8 — R87 0R — VCC_0V8; C155 100nF, C156 100nF

VCC1V8	VCC1V8
VCC0V8	VCC0V8
VCC1V8_DOVDD	VCC1V8_DOVDD
VCC1V2_DVDD	VCC1V2_DVDD
VCC2V8_AVDD	VCC2V8_AVDD
GND	GND

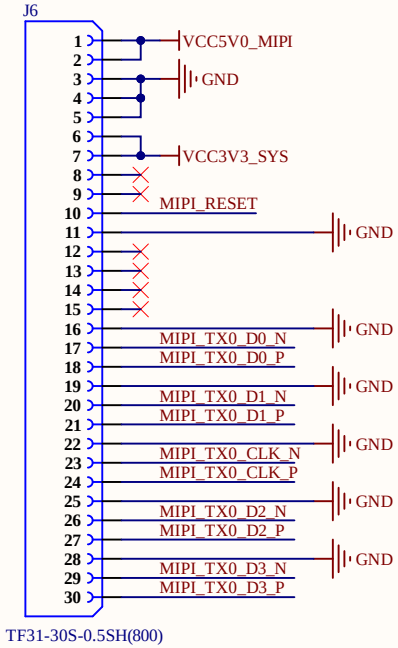
eSpark Consulting Group

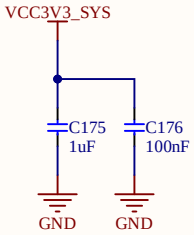
Title: MIPI-CSI_RXX_INTERFACE.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page Num:	22
Drawn By: Huzaifa Ali	Page Total:	27

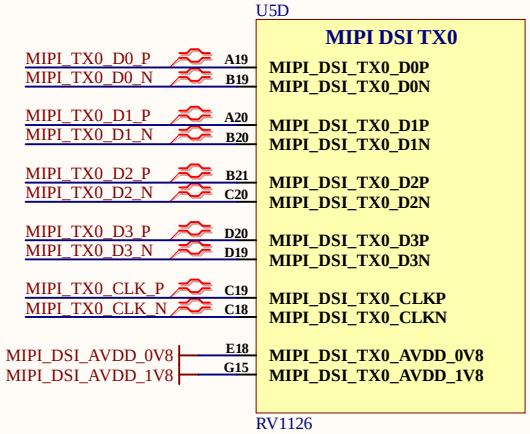
MIPI TO HDMI_MODULE_INTERFACE

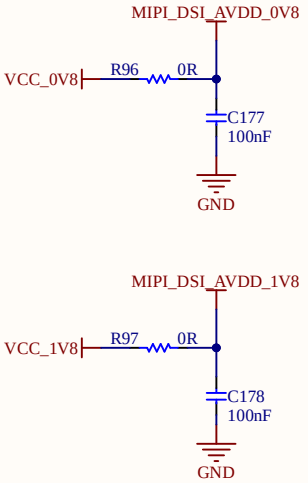


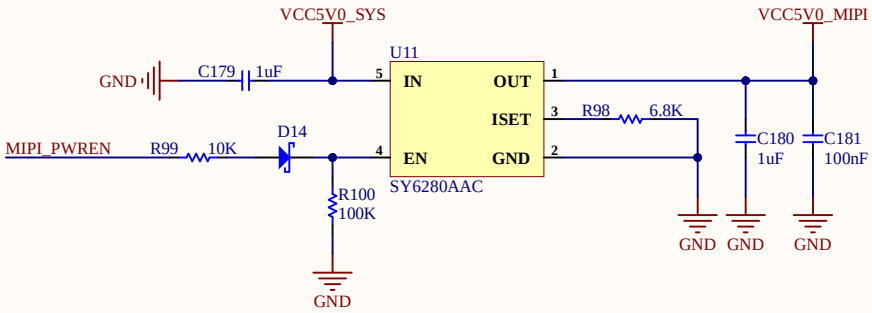
1	5V	5V 供电
2	5V	
3	GND	地
4	GND	
5	GND	
6	VDD33	3.3V 供电
7	VDD33	
8	LCD 控制信号	不需要接
9		
10	LCD-RST	接主控的 IO 拉高开显示，拉低关显示
11	GND	地
12	触摸屏信号	不需要接
13		
14		
15		
16	GND	地
17	MIPI-D0N	
18	MIPI-D0P	
19	GND	地
20	MIPI-D1N	
21	MIPI-D1P	
22	GND	地
23	MIPI-CLKN	
24	MIPI-CLKP	
25	GND	地
26	MIPI-D2N	
27	MIPI-D2P	
28	GND	地
29	MIPI-D3N	
30	MIPI-D3P	

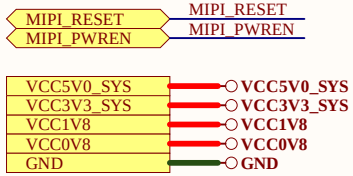








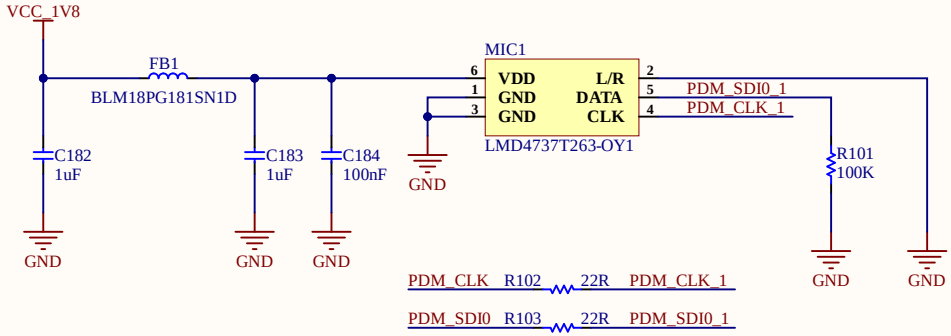


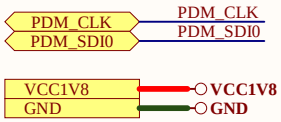


eSpark Consulting Group

Title: MIPI TO HDMI_MODULE.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page Num:	23
Drawn By: Huzaifa Ali	Page Total:	27

DIGITAL_MEMS_MIC_INTERFACE



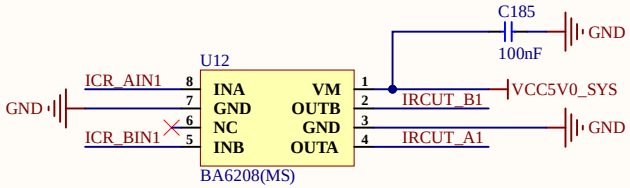


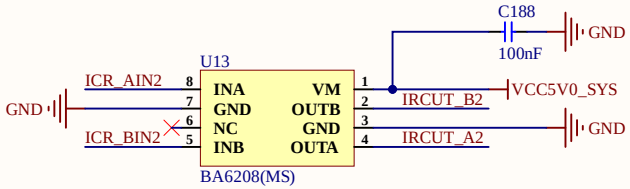
eSpark Consulting Group

Title: DIGITAL_MEMS_INTERFACE.SchDoc	
File: PGS-System.PrjPcb	
Date: 09/08/2024	Page Num: 24
Drawn By: Huzaifa Ali	Page Total: 27

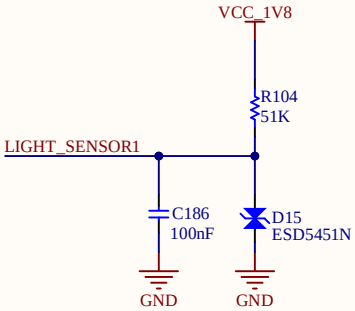
IR-CUT_LIGHT SENSOR_INTERFACE

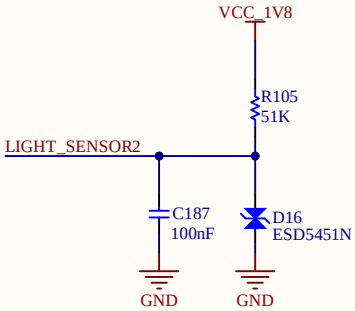
IRCUT_DRIVER

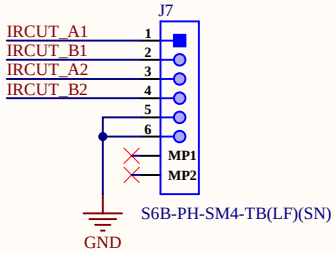


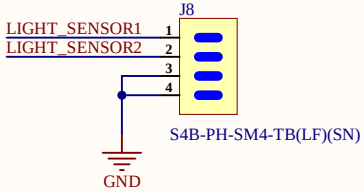


LIGHT_SENSOR











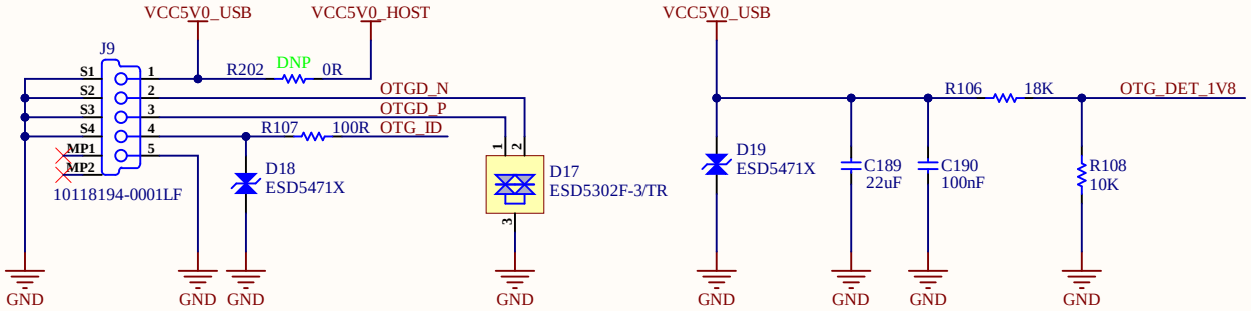


eSpark Consulting Group

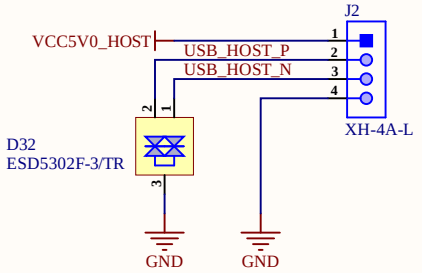
Title: IR-CUT_LIGHT SENSOR_SHEET.SchDoc	
File: PGS-System.PrjPcb	
Date: 09/08/2024	Page Num: 25
Drawn By: Huzaifa Ali	Page Total: 27

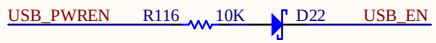
USB2.0_OTG_HOST

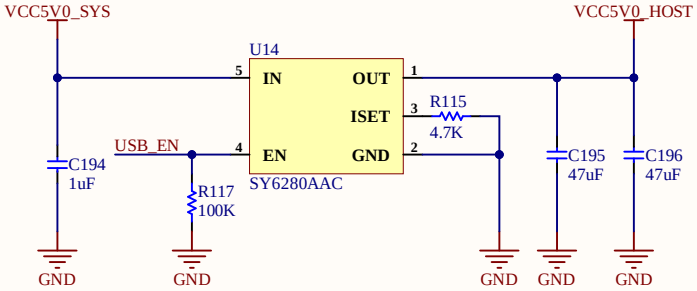
USB OTG 2.0

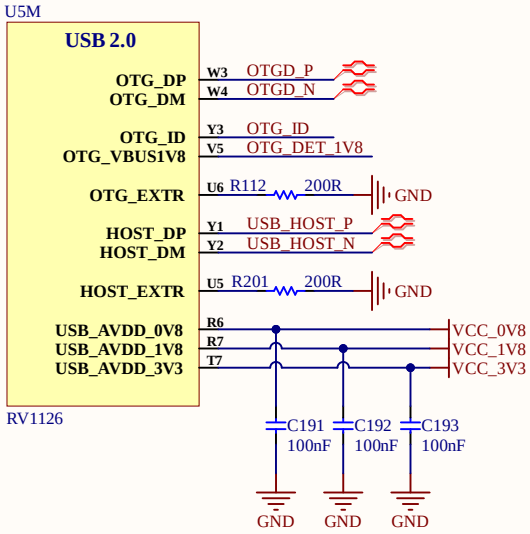


USB HOST 2.0

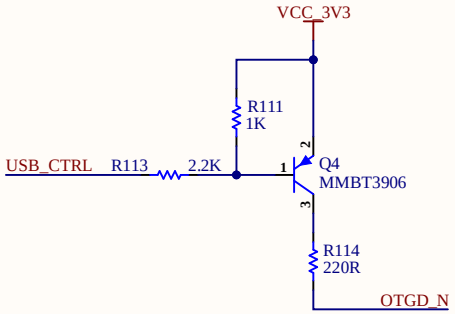




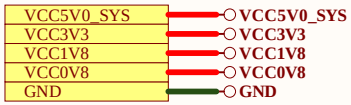




Note:
This circuit is used to improve usb compatibility





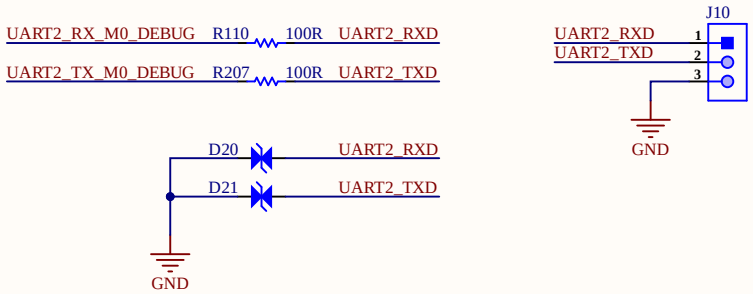


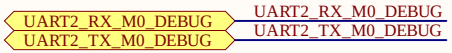
eSpark Consulting Group

Title: USB2.0_DEBUG_UART.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 26
Drawn By: Huzaifa Ali	Page	Total: 27

USB2.0_OTG_HOST

Debug UART







eSpark Consulting Group

Title: DEBUG_UART_INTERFACE.SchDoc		
File: PGS-System.PrjPcb		
Date: 09/08/2024	Page	Num: 27
Drawn By: Huzaifa Ali	Page	Total: 27